

Class 10: Structural Bioinformatics 1

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The main repository of high-resolution structural data on biomolecules is called the **Protein Data Bank** (PDB).

PDB statistics

What is in the PDB in terms of molecule type and structure determination method?

Read a CSV file of current PDB stats obtained from <https://www.rcsb.org/stats/summary>

```
pdb <- read.csv("Data Export Summary.csv")
pdb
```

	Molecular.Type	X.ray	EM	NMR	Integrative	Multiple.methods
1	Protein (only)	180,758	23,111	12,813	348	229
2	Protein/Oligosaccharide	10,488	3,741	34	8	11
3	Protein/NA	9,205	6,751	287	26	8
4	Nucleic acid (only)	3,154	250	1,578	3	15
5	Other	178	27	35	4	0
6	Oligosaccharide (only)	11	0	6	0	1

	Neutron	Other	Total
1	84	32	217,375
2	1	0	14,283
3	0	0	16,277
4	3	1	5,004
5	0	0	244
6	0	4	22

```
pdb$X.ray
```

```
[1] "180,758" "10,488" "9,205" "3,154" "178" "11"
```

This print out above `pdb$X.ray` is “character” not “numeric”. Therefore I can’t do math with it. We need to fix this...

Two functions that can help here are `sub()` and `as.numeric()`

```
# We want to get rid (or sub out) commas:
x <- pdb$X.ray
tmp <- sub(",", "", x)
sum(as.numeric(tmp))
```

```
[1] 203794
```

We could make a function to do this:

```
rm.comma <- function(x) {
  tmp <- sub(",", "", x)
  sum(as.numeric(tmp))
}
```

```
n.tot <- rm.comma(pdb$Total)
n.xray <- rm.comma(pdb$X.ray)
n.em <- rm.comma(pdb$EM)

n.xray/n.tot * 100
```

```
[1] 80.48577
```

```
n.em/n.tot * 100
```

```
[1] 13.38046
```

Q1. What percentage of structures in the PDB are solved by X-Ray and Electron Microscopy?

80.49% of structures in the PDB are solved by X-Ray and 13.38% by Electron Microscopy.

We could also use a different import function for this CSV that speaks American (i.e. deals with commas in numbers in a comma separated value file)

```
library(readr)

pdb <- read_csv("Data Export Summary.csv")
```

```
Rows: 6 Columns: 9
```

```
-- Column specification -----
```

```
Delimiter: ","
```

```
chr (1): Molecular Type
```

```
dbl (4): Integrative, Multiple methods, Neutron, Other
```

```
num (4): X-ray, EM, NMR, Total
```

```
i Use `spec()` to retrieve the full column specification for this data.
```

```
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

Q2. How many total protein structures are there in the dataset? What proportion of structures in the PDB are protein?

```
pdb$Total[1]
```

```
[1] 217375
```

The total number of protein sequences in UniProt is 202,554,314.

```
217375 / 202554314 * 100
```

```
[1] 0.1073169
```

Key-point: We have a very, very small structural coverage of known proteins (~0.1%). Most structures we know about (~80%) come from one method (X-ray crystallography).

Q3: Type HIV in the PDB website search box on the home page and determine how many HIV-1 protease structures are in the current PDB?

There are 783 HIV-1 protease structures in the current PDB.

Visualizing PDB data with Mol-star

Main stand alone web version with all features is at <https://molstar.org/viewer/>.

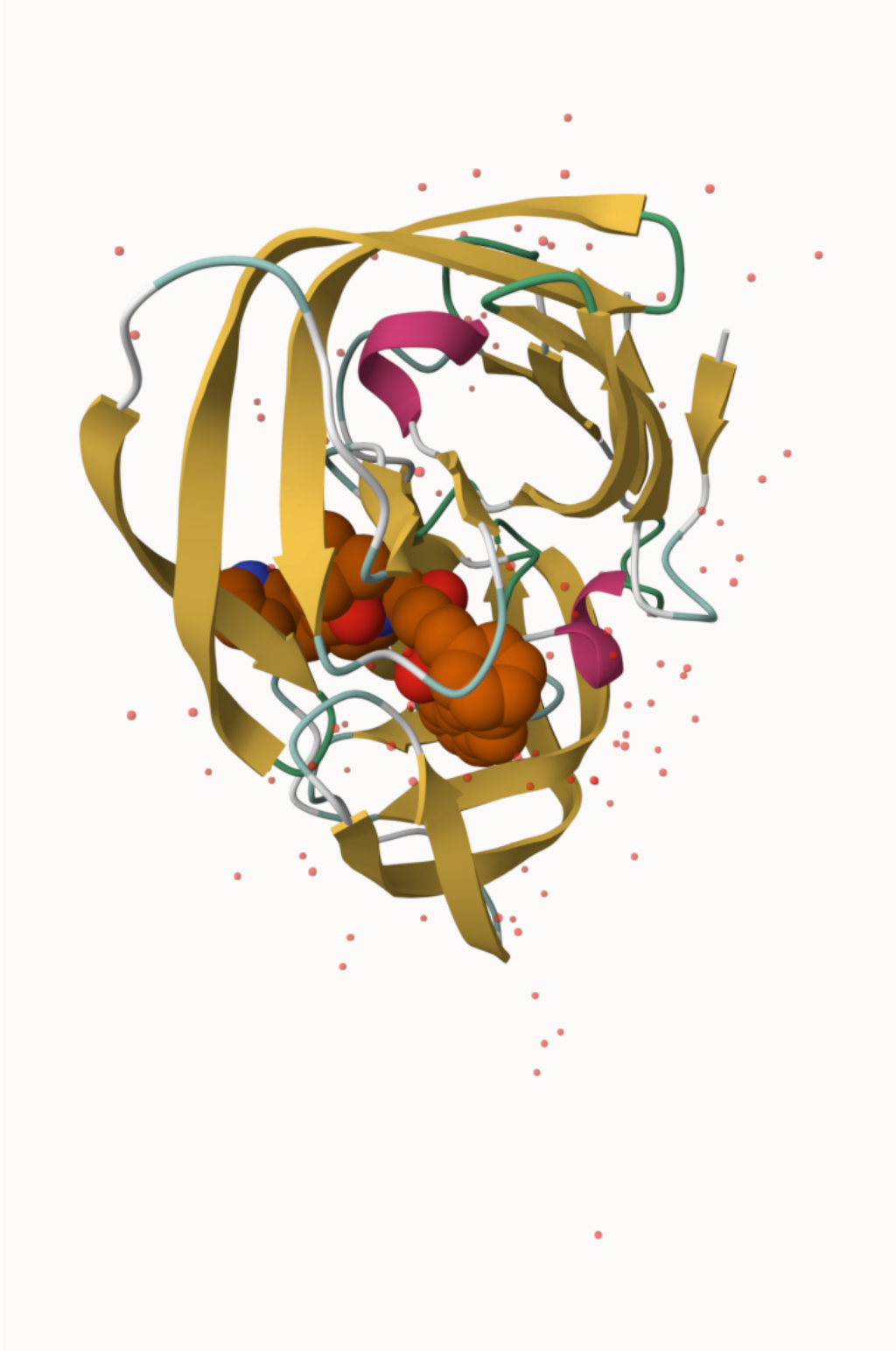


Figure 1: The HIV-protease enzyme is a homodimer with two chains.

Q4: Water molecules normally have 3 atoms. Why do we see just one atom per water molecule in this structure?

The hydrogen atoms are usually not visible, especially at 2.00 Å resolution, so each water molecule is shown only by its oxygen atom.

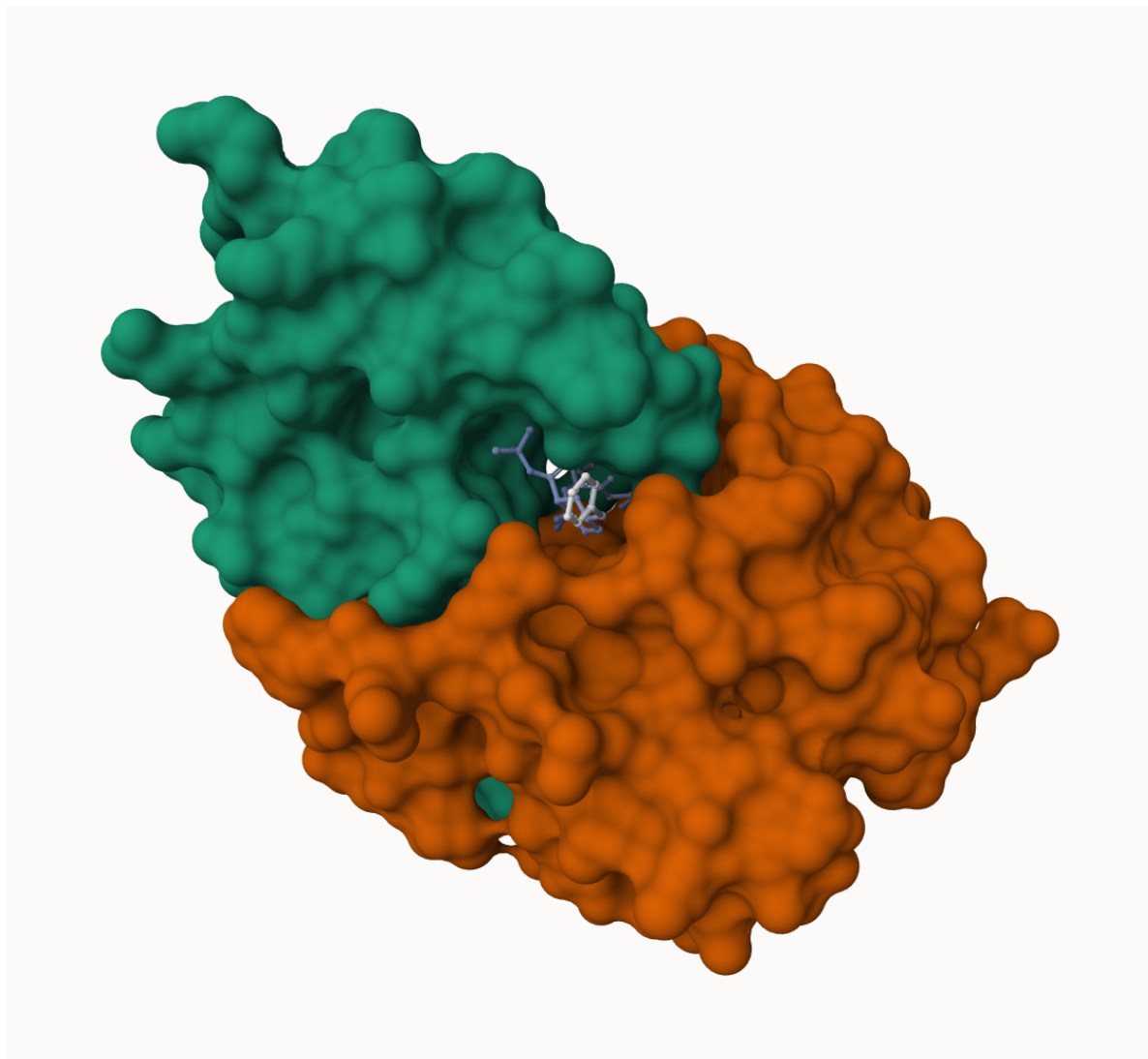


Figure 2: Surface display showing the binding cleft site for the inhibitor (drug) molecule.

Q5: There is a critical “conserved” water molecule in the binding site. Can you identify this water molecule? What residue number does this water molecule have?

The critical “conserved” water molecule in the binding site is residue 308. It bridges interactions within the active site.

Q6: Generate and save a figure clearly showing the two distinct chains of HIV-protease along with the ligand. You might also consider showing the catalytic residues ASP 25 in each chain and the critical water (we recommend “Ball & Stick” for these side-chains). Add this figure to your Quarto document.

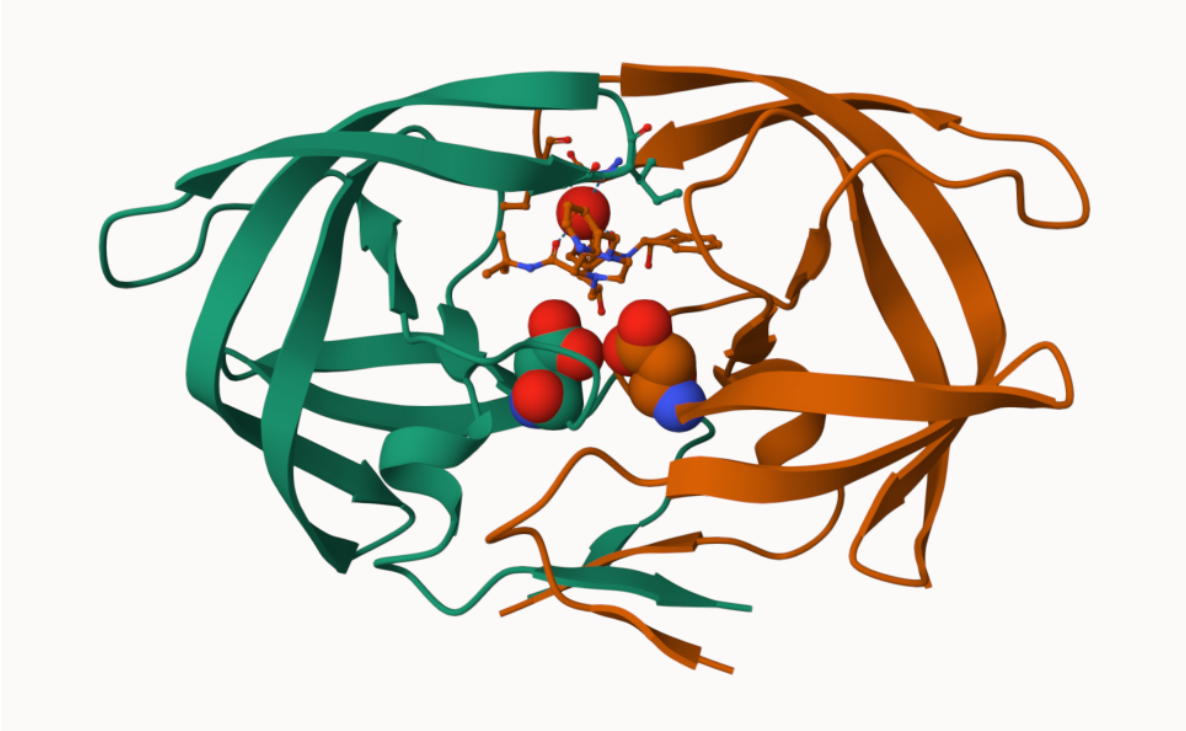


Figure 3: Spacefill display of catalytic ASP25 amino acids and key binding site water molecule.

Getting started with the Bio3D package

Bio3D is an R package from CRAN for structural bioinformatics.

```
library(bio3d)
pdb <- read.pdb("1hsg")
```

Note: Accessing on-line PDB file

```
pdb
```

```
Call: read.pdb(file = "1hsg")
```

```
Total Models#: 1
```

```
Total Atoms#: 1686, XYZs#: 5058 Chains#: 2 (values: A B)
```

```
Protein Atoms#: 1514 (residues/Calpha atoms#: 198)
```

```
Nucleic acid Atoms#: 0 (residues/phosphate atoms#: 0)
```

```
Non-protein/nucleic Atoms#: 172 (residues: 128)
```

```
Non-protein/nucleic resid values: [ HOH (127), MK1 (1) ]
```

```
Protein sequence:
```

```
PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGIGGFIKVRQYD  
QILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNFPQITLWQRPLVTIKIGGQLKE  
ALLDTGADDTVLEEMSLPGRWKPKMIGGIGGFIKVRQYDQILIEICGHKAIGTVLVGPTP  
VNIIGRNLLTQIGCTLNF
```

```
+ attr: atom, xyz, seqres, helix, sheet,  
      calpha, remark, call
```

Q7: How many amino acid residues are there in this pdb object?

There are 198 residues in this pdb object.

Q8: Name one of the two non-protein residues?

One of the two non-protein residues is water (HOH).

Q9: How many protein chains are in this structure?

There are 2 protein chains in this structure.

```
attributes(pdb)
```

```
$names
```

```
[1] "atom" "xyz" "seqres" "helix" "sheet" "calpha" "remark" "call"
```

```
$class
```

```
[1] "pdb" "sse"
```

```
head(pdb$atom)
```


	type	eleno	elety	alt	resid	chain	resno	insert	x	y	z	o	b
1	ATOM	1	N	<NA>	PRO	A	1	<NA>	29.361	39.686	5.862	1	38.10
2	ATOM	2	CA	<NA>	PRO	A	1	<NA>	30.307	38.663	5.319	1	40.62
3	ATOM	3	C	<NA>	PRO	A	1	<NA>	29.760	38.071	4.022	1	42.64
4	ATOM	4	O	<NA>	PRO	A	1	<NA>	28.600	38.302	3.676	1	43.40
5	ATOM	5	CB	<NA>	PRO	A	1	<NA>	30.508	37.541	6.342	1	37.87
6	ATOM	6	CG	<NA>	PRO	A	1	<NA>	29.296	37.591	7.162	1	38.40

	segid	elesy	charge
1	<NA>	N	<NA>
2	<NA>	C	<NA>
3	<NA>	C	<NA>
4	<NA>	O	<NA>
5	<NA>	C	<NA>
6	<NA>	C	<NA>

There are lots of functions that can work with these `pdb` objects:

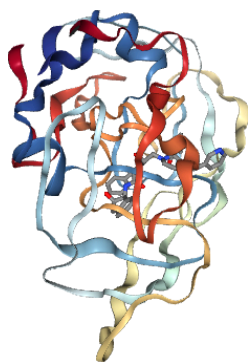
```
head(pdbseq(pdb))
```

```
  1    2    3    4    5    6
"P" "Q" "I" "T" "L" "W"
```

```
library(bio3dview)
```

```
view.pdb(pdb)
```

```
Warning in basename(x): expanded path length 136572 would be too long for
ATOM      1  N   PRO A   1       29.361  39.686   5.862   1.00 38.10           N
ATOM      2  CA  PRO A   1       30.307  38.663   5.319   1.00 40.62           C
ATOM      3  C   PRO A   1       29.760  38.071   4.022   1.00 42.64           C
ATOM      4  O   PRO A   1       28.600  38.302   3.676   1.00 43.40           O
ATOM      5  CB  PRO A   1       30.508  37.541   6.342   1.00 37.87           C
ATOM      6  CG  PRO A   1       29.296  37.591   7.162   1.00 38.40           C
ATOM      7  CD  PRO A   1       28.778  39.015   7.019   1.00 38.74           C
ATOM      8  N   GLN A   2       30.607  37.334   3.305   1.00 41.76           N
ATOM      9  CA  GLN A   2       30.158  36.492   2.199   1.00 41.30           C
ATOM     10  C   GLN A   2       30.298  35.041   2.643   1.00 41.38           C
ATOM     11  O   GLN A   2       31.401  34.494   2.763   1.00 43.09           O
ATOM     12  CB  GLN A   2       30.970  36.738   0.926   1.0 [... truncated]
```

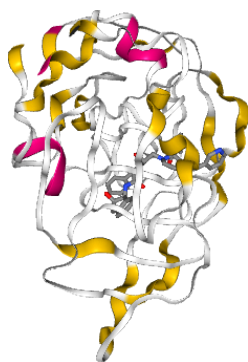


Let's try a custom view

```
view.pdb(pdb,  
         colorScheme="sse",  
         backgroundColor = "black")
```

Warning in basename(x): expanded path length 136572 would be too long for

ATOM	1	N	PRO	A	1	29.361	39.686	5.862	1.00	38.10	N
ATOM	2	CA	PRO	A	1	30.307	38.663	5.319	1.00	40.62	C
ATOM	3	C	PRO	A	1	29.760	38.071	4.022	1.00	42.64	C
ATOM	4	O	PRO	A	1	28.600	38.302	3.676	1.00	43.40	O
ATOM	5	CB	PRO	A	1	30.508	37.541	6.342	1.00	37.87	C
ATOM	6	CG	PRO	A	1	29.296	37.591	7.162	1.00	38.40	C
ATOM	7	CD	PRO	A	1	28.778	39.015	7.019	1.00	38.74	C
ATOM	8	N	GLN	A	2	30.607	37.334	3.305	1.00	41.76	N
ATOM	9	CA	GLN	A	2	30.158	36.492	2.199	1.00	41.30	C
ATOM	10	C	GLN	A	2	30.298	35.041	2.643	1.00	41.38	C
ATOM	11	O	GLN	A	2	31.401	34.494	2.763	1.00	43.09	O
ATOM	12	CB	GLN	A	2	30.970	36.738	0.926	1.0	[... truncated]	



Q. Create a custom view of HIV-Pr highlighting the active site ASP residues (resno=25), the two chains (in your choice of colors), and the ligand all on a custom color background?

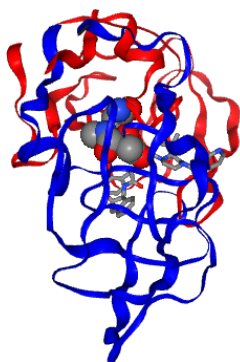
```
library(bio3dview)
library(NGLViewerR)

active.site <- atom.select(pdb, resno = 25)

view.pdb(pdb,
  cols = c("red", "blue"),
  highlight = active.site,
  highlight.style = "spacefill",
  backgroundColor = "pink") |>
  setRock()
```

Warning in basename(x): expanded path length 136572 would be too long for

ATOM	1	N	PRO	A	1	29.361	39.686	5.862	1.00	38.10	N
ATOM	2	CA	PRO	A	1	30.307	38.663	5.319	1.00	40.62	C
ATOM	3	C	PRO	A	1	29.760	38.071	4.022	1.00	42.64	C
ATOM	4	O	PRO	A	1	28.600	38.302	3.676	1.00	43.40	O
ATOM	5	CB	PRO	A	1	30.508	37.541	6.342	1.00	37.87	C
ATOM	6	CG	PRO	A	1	29.296	37.591	7.162	1.00	38.40	C
ATOM	7	CD	PRO	A	1	28.778	39.015	7.019	1.00	38.74	C
ATOM	8	N	GLN	A	2	30.607	37.334	3.305	1.00	41.76	N
ATOM	9	CA	GLN	A	2	30.158	36.492	2.199	1.00	41.30	C
ATOM	10	C	GLN	A	2	30.298	35.041	2.643	1.00	41.38	C
ATOM	11	O	GLN	A	2	31.401	34.494	2.763	1.00	43.09	O
ATOM	12	CB	GLN	A	2	30.970	36.738	0.926	1.0	[... truncated]	



Predict the flexibility of a given structure

Let's do a Normal Mode Analysis (NMA) to predict the flexibility of a given `pdb` object:

```
adk <- read.pdb("6s36")
```

Note: Accessing on-line PDB file
PDB has ALT records, taking A only, `rm.alt=TRUE`

A quick structure summary

```
adk
```

```
Call: read.pdb(file = "6s36")
```

```
Total Models#: 1
```

```
Total Atoms#: 1898, XYZs#: 5694 Chains#: 1 (values: A)
```

```
Protein Atoms#: 1654 (residues/Calpha atoms#: 214)
```

```
Nucleic acid Atoms#: 0 (residues/phosphate atoms#: 0)
```

```
Non-protein/nucleic Atoms#: 244 (residues: 244)
```

```
Non-protein/nucleic resid values: [ CL (3), HOH (238), MG (2), NA (1) ]
```

```
Protein sequence:
```

```
MRILLGAPGAGKGTQAQFIMEKYGIPQISTGDMRLRAAVKSGSELGKQAKDIMDAGKLV  
DELVIALVKERIAQEDCRNGFLLDGFPRTPQADAMKEAGINVDYVLEFDVPDELIVDKI  
VGRRVHAPSGRVYHVKFNPPKVEGKDDVTGEELTTRKDDQEETVRKRLVEYHQMTAPLIG  
YYSKEAEAGNTKYAKVDGTPVAEVRADLEKILG
```

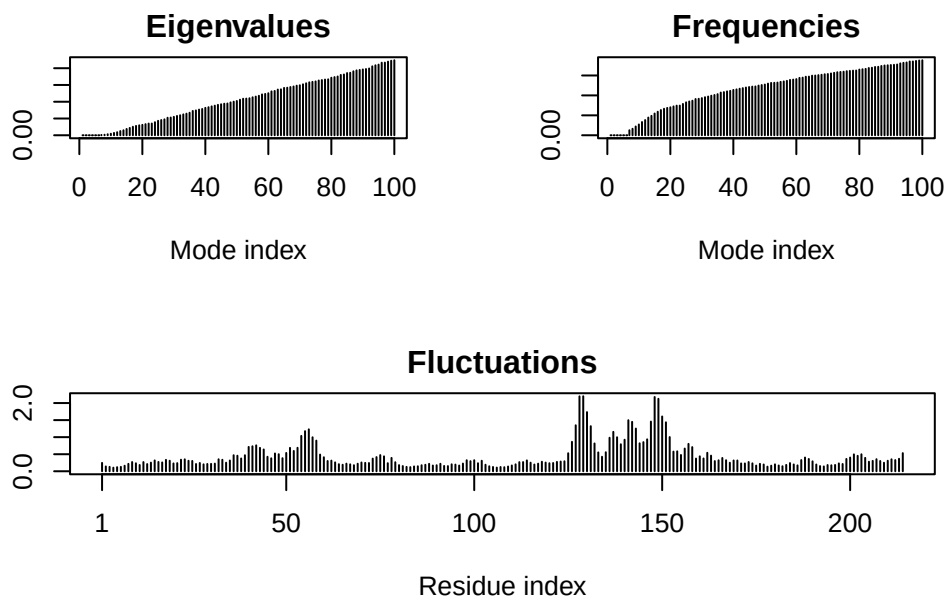
```
+ attr: atom, xyz, seqres, helix, sheet,  
      calpha, remark, call
```

```
m <- nma(adk)
```

```
Building Hessian... Done in 0.08 seconds.
```

```
Diagonalizing Hessian... Done in 0.526 seconds.
```

```
plot(m)
```

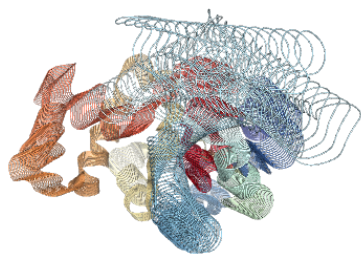


View the results with an interactive structure view:

```
view.nma(m)
```

Warning in basename(x): expanded path length 590110 would be too long for

```
MODEL      1
ATOM      1  CA  ALA      1    -19.326   9.682  18.832   1.00   0.00
ATOM      2  CA  ALA      2    -16.894   6.998  17.674   1.00   0.00
ATOM      3  CA  ALA      3    -18.316   5.203  14.643   1.00   0.00
ATOM      4  CA  ALA      4    -17.091   2.435  12.363   1.00   0.00
ATOM      5  CA  ALA      5    -18.775   2.508   8.937   1.00   0.00
ATOM      6  CA  ALA      6    -18.464  -0.698   6.902   1.00   0.00
ATOM      7  CA  ALA      7    -19.758  -1.475   3.453   1.00   0.00
ATOM      8  CA  ALA      8    -19.074  -3.493   0.350   1.00   0.00
ATOM      9  CA  ALA      9    -17.159  -1.879  -2.535   1.00   0.00
ATOM     10  CA  ALA     10    -19.711  -0.041  -4.657   1.00   0.00
ATOM     11  CA  ALA     11    -22.527   0.023  -2.105   1.00   0.00
ATOM     12  CA  ALA     12    -22.175   3.7 [... truncated]
```

Write out the results for viewing in Mol-star:

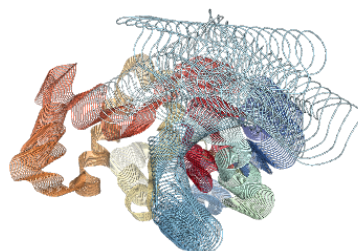
```
mktrj(m, file = "nma.pdb")
```

You can view quickly here or open the file made previously in Mol-star.

```
view.nma(m, pdb=adk)
```

Warning in basename(x): expanded path length 590110 would be too long for

```
MODEL      1
ATOM       2  CA  MET  A   1    -19.326   9.682  18.832   1.00  36.10      C
ATOM      11  CA AARG  A   2    -16.894   6.998  17.674   0.50  26.59      C
ATOM      32  CA  ILE  A   3    -18.316   5.203  14.643   1.00  21.76      C
ATOM      40  CA  ILE  A   4    -17.091   2.435  12.363   1.00  20.57      C
ATOM      48  CA  LEU  A   5    -18.775   2.508   8.937   1.00  19.08      C
ATOM      56  CA  LEU  A   6    -18.464  -0.698   6.902   1.00  24.62      C
ATOM      64  CA  GLY  A   7    -19.758  -1.475   3.453   1.00  25.04      C
ATOM      68  CA  ALA  A   8    -19.074  -3.493   0.350   1.00  26.93      C
ATOM      73  CA  PRO  A   9    -17.159  -1.879  -2.535   1.00  27.11      C
ATOM      80  CA  GLY  A  10    -19.711  -0.041  -4.657   1.00  30.50      C
ATOM      84  CA  ALA  A  11    -22.527   0.023  -2.105   1.00  31.59      C
ATOM      89  CA  GLY  A  12    -22.175   3.7 [... truncated]
```



Comparative analysis of the ADK family

```
# Install packages in the R console NOT your Rmd/Quarto file
# install.packages("bio3d")
# install.packages("NGLVieweR")

#install.packages("remotes")
#remotes::install_github("bioboot/bio3dview")

#install.packages("BiocManager")
#BiocManager::install("msa")
```

Q10. Which of the packages above is found only on BioConductor and not CRAN?

The package only found on BioConductor is “msa”.

Q11. Which of the above packages is not found on BioConductor or CRAN?

The package not found on BioConductor or CRAN is bio3dview.(It is installed from GitHub).

Q12. True or False? Functions from the pak package can be used to install packages from GitHub and BitBucket?

True.

Our first step is find a sequence for this family. We will use the database iD “1ake_A” here:

```
id <- "1ake_A"
aa <- get.seq(id)
```

Warning in get.seq(id): Removing existing file: seqs.fasta

Fetching... Please wait. Done.

aa

	1	.	.	.	.	.	60
pdb 1AKE A	MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMRLRAAVKSGSELGKQAKDIMDAGKLV						
	1	.	.	.	.	.	60
	61	.	.	.	.	.	120

```

pdb|1AKE|A    DELVIALVKERIAQEDCRNGFLLDGFPR TIPQADAMKEAGINVDYVLEFDVPDELIVDRI
              61          .          .          .          .          .          120

              121          .          .          .          .          .          180
pdb|1AKE|A    VGRRVHAPSGRVYHVKNPPKVEGKDDVTGEELTTRKDDQEETVRKRLVEYHQMTAPLIG
              121          .          .          .          .          .          180

              181          .          .          .          214
pdb|1AKE|A    YYSKEAEAGNTKYAKVDGTPVAEVRADLEKILG
              181          .          .          .          214

```

Call:

```
read.fasta(file = outfile)
```

Class:

```
fasta
```

Alignment dimensions:

```
1 sequence rows; 214 position columns (214 non-gap, 0 gap)
```

```
+ attr: id, ali, call
```

Q13. How many amino acids are in this sequence, i.e. how long is this sequence?

There are 214 amino acids in this sequence.

Search for related sequences in the database

```
blast <- blast.pdb(aa)
```

Searching ... please wait (updates every 5 seconds) RID = ZTV3HC8X016

.....

Reporting 96 hits

```
hits <- plot(blast)
```

```

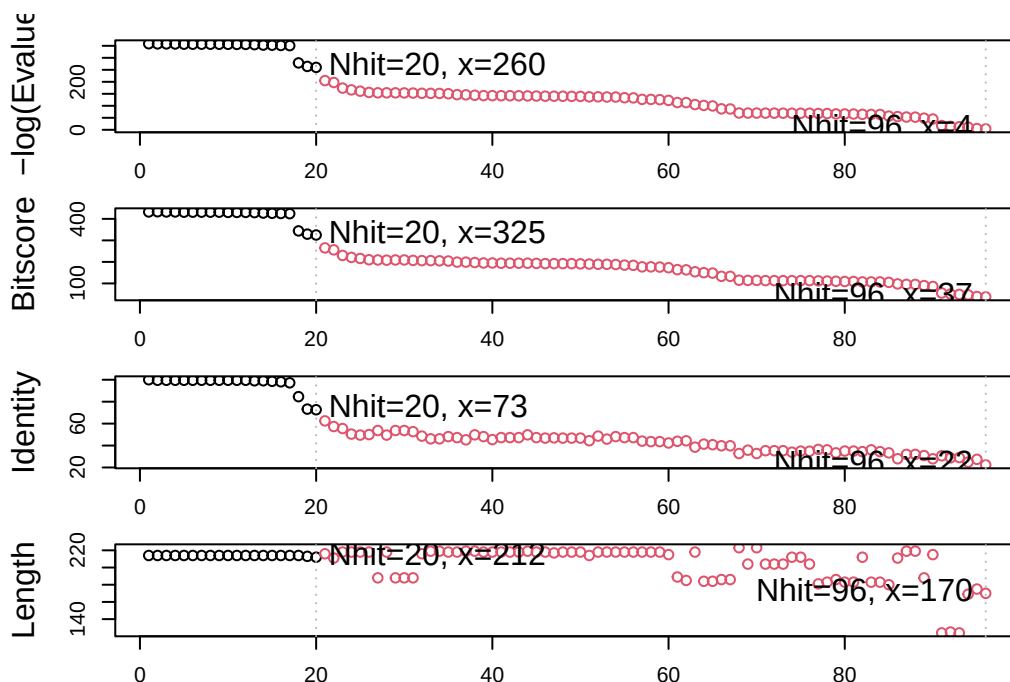
* Possible cutoff values:    260 3
    Yielding Nhits:         20 96

```

```

* Chosen cutoff value of:    260
    Yielding Nhits:          20

```



```
hits$pdb.id
```

```
[1] "1AKE_A" "8BQF_A" "4X8M_A" "6S36_A" "9R6U_A" "9R71_A" "8Q2B_A" "8RJ9_A"
[9] "6RZE_A" "4X8H_A" "3HPR_A" "1E4V_A" "5EJE_A" "1E4Y_A" "3X2S_A" "6HAP_A"
[17] "6HAM_A" "8PVW_A" "4K46_A" "4NP6_A"
```

```
# Download related PDB files
files <- get.pdb(hits$pdb.id, path="pdb", split=TRUE, gzip=TRUE)
```

```
Warning in get.pdb(hits$pdb.id, path = "pdb", split = TRUE, gzip = TRUE):
pdb/1AKE.pdb.gz exists. Skipping download
```

```
Warning in get.pdb(hits$pdb.id, path = "pdb", split = TRUE, gzip = TRUE):
pdb/8BQF.pdb.gz exists. Skipping download
```

```
Warning in get.pdb(hits$pdb.id, path = "pdb", split = TRUE, gzip = TRUE):
pdb/4X8M.pdb.gz exists. Skipping download
```

```
Warning in get.pdb(hits$pdb.id, path = "pdb", split = TRUE, gzip = TRUE):
pdb/6S36.pdb.gz exists. Skipping download
```

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/9R6U.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/9R71.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/8Q2B.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/8RJ9.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/6RZE.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/4X8H.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/3HPR.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/1E4V.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/5EJE.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/1E4Y.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/3X2S.pdb.gz exists. Skipping download

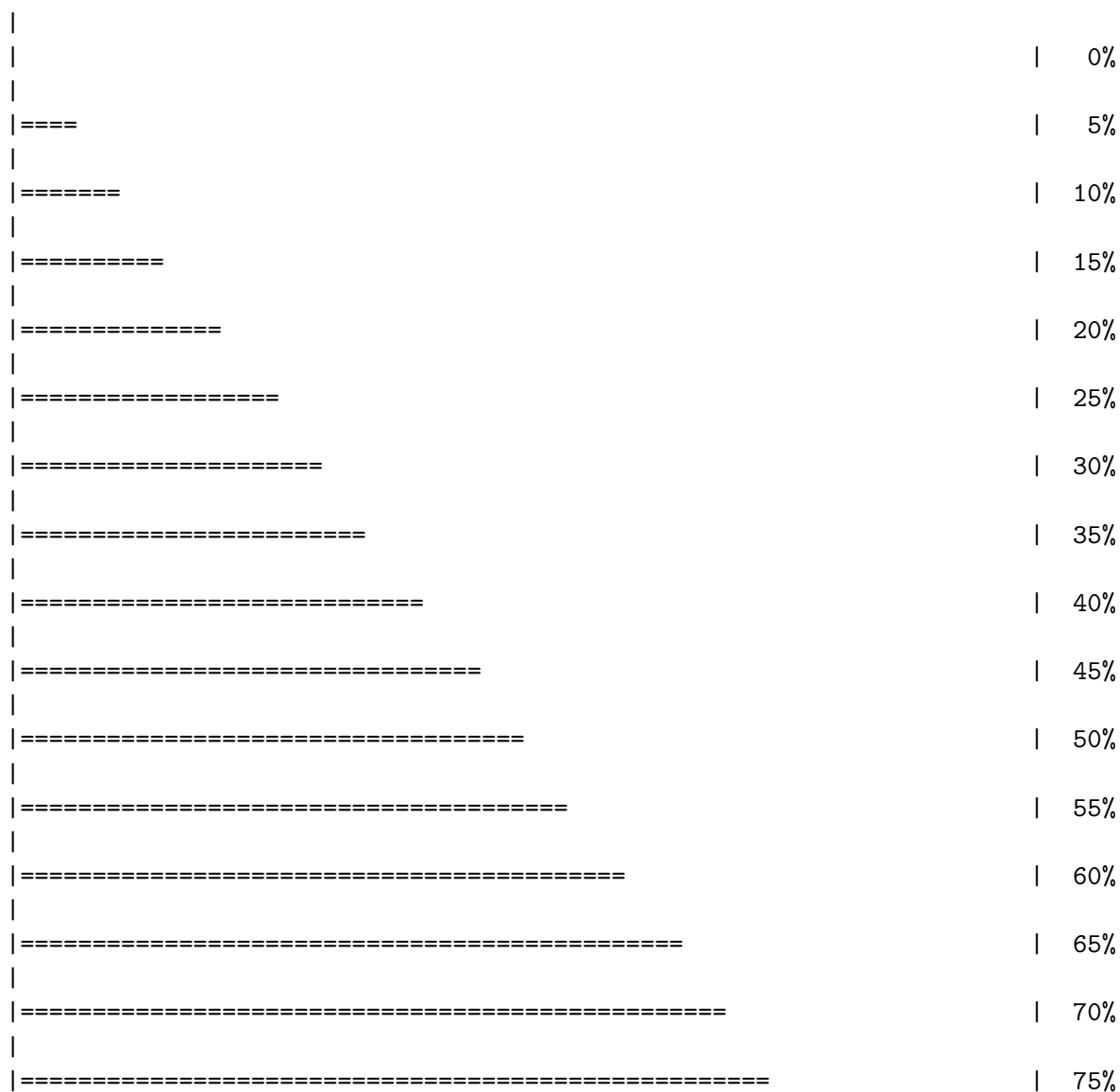
Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/6HAP.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/6HAM.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/8PVW.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/4K46.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/4NP6.pdb.gz exists. Skipping download




```

|
|=====| 80%
|
|=====| 85%
|
|=====| 90%
|
|=====| 95%
|
|=====| 100%

```

Align and superpose all these ADK structures

```
# Align related PDBs
pdbbs <- pdbaln(files, fit = TRUE, exefile="msa")
```

Reading PDB files:

```

pdbbs/split_chain/1AKE_A.pdb
pdbbs/split_chain/8BQF_A.pdb
pdbbs/split_chain/4X8M_A.pdb
pdbbs/split_chain/6S36_A.pdb
pdbbs/split_chain/9R6U_A.pdb
pdbbs/split_chain/9R71_A.pdb
pdbbs/split_chain/8Q2B_A.pdb
pdbbs/split_chain/8RJ9_A.pdb
pdbbs/split_chain/6RZE_A.pdb
pdbbs/split_chain/4X8H_A.pdb
pdbbs/split_chain/3HPR_A.pdb
pdbbs/split_chain/1E4V_A.pdb
pdbbs/split_chain/5EJE_A.pdb
pdbbs/split_chain/1E4Y_A.pdb
pdbbs/split_chain/3X2S_A.pdb
pdbbs/split_chain/6HAP_A.pdb
pdbbs/split_chain/6HAM_A.pdb
pdbbs/split_chain/8PVW_A.pdb
pdbbs/split_chain/4K46_A.pdb
pdbbs/split_chain/4NP6_A.pdb
  PDB has ALT records, taking A only, rm.alt=TRUE
.   PDB has ALT records, taking A only, rm.alt=TRUE
..  PDB has ALT records, taking A only, rm.alt=TRUE
.   PDB has ALT records, taking A only, rm.alt=TRUE
.   PDB has ALT records, taking A only, rm.alt=TRUE

```

```

.   PDB has ALT records, taking A only, rm.alt=TRUE
.   PDB has ALT records, taking A only, rm.alt=TRUE
.   PDB has ALT records, taking A only, rm.alt=TRUE
..  PDB has ALT records, taking A only, rm.alt=TRUE
..  PDB has ALT records, taking A only, rm.alt=TRUE
.... PDB has ALT records, taking A only, rm.alt=TRUE
.   PDB has ALT records, taking A only, rm.alt=TRUE
.   PDB has ALT records, taking A only, rm.alt=TRUE
..

```

Extracting sequences

```

pdb/seq: 1   name: pdbc/split_chain/1AKE_A.pdb
           PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 2   name: pdbc/split_chain/8BQF_A.pdb
           PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 3   name: pdbc/split_chain/4X8M_A.pdb
pdb/seq: 4   name: pdbc/split_chain/6S36_A.pdb
           PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 5   name: pdbc/split_chain/9R6U_A.pdb
           PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 6   name: pdbc/split_chain/9R71_A.pdb
           PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 7   name: pdbc/split_chain/8Q2B_A.pdb
           PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 8   name: pdbc/split_chain/8RJ9_A.pdb
           PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 9   name: pdbc/split_chain/6RZE_A.pdb
           PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 10  name: pdbc/split_chain/4X8H_A.pdb
pdb/seq: 11  name: pdbc/split_chain/3HPR_A.pdb
           PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 12  name: pdbc/split_chain/1E4V_A.pdb
pdb/seq: 13  name: pdbc/split_chain/5EJE_A.pdb
           PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 14  name: pdbc/split_chain/1E4Y_A.pdb
pdb/seq: 15  name: pdbc/split_chain/3X2S_A.pdb
pdb/seq: 16  name: pdbc/split_chain/6HAP_A.pdb
pdb/seq: 17  name: pdbc/split_chain/6HAM_A.pdb
           PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 18  name: pdbc/split_chain/8PVW_A.pdb
           PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 19  name: pdbc/split_chain/4K46_A.pdb

```

PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 20 name: pdbs/split_chain/4NP6_A.pdb

pdbs

```

[Truncated_Name:1] 1AKE_A.pdb      1      .      .      .      40
[Truncated_Name:2] 8BQF_A.pdb      --MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMRLAA
[Truncated_Name:3] 4X8M_A.pdb      --MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMRLAA
[Truncated_Name:4] 6S36_A.pdb      --MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMRLAA
[Truncated_Name:5] 9R6U_A.pdb      --MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMRLAA
[Truncated_Name:6] 9R71_A.pdb      --MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMRLAA
[Truncated_Name:7] 8Q2B_A.pdb      --MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMRLAA
[Truncated_Name:8] 8RJ9_A.pdb      --MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMRLAA
[Truncated_Name:9] 6RZE_A.pdb      --MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMRLAA
[Truncated_Name:10] 4X8H_A.pdb     --MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMRLAA
[Truncated_Name:11] 3HPR_A.pdb     --MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMRLAA
[Truncated_Name:12] 1E4V_A.pdb     --MRIILLGAPVAGKGTQAQFIMEKYGIPQISTGDMRLAA
[Truncated_Name:13] 5EJE_A.pdb     --MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMRLAA
[Truncated_Name:14] 1E4Y_A.pdb     --MRIILLGALVAGKGTQAQFIMEKYGIPQISTGDMRLAA
[Truncated_Name:15] 3X2S_A.pdb     --MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMRLAA
[Truncated_Name:16] 6HAP_A.pdb     --MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMRLAA
[Truncated_Name:17] 6HAM_A.pdb     --MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMRLAA
[Truncated_Name:18] 8PVW_A.pdb     --MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMRLAA
[Truncated_Name:19] 4K46_A.pdb     --MRIILLGAPGAGKGTQAQFIMAKFGIPQISTGDMRLAA
[Truncated_Name:20] 4NP6_A.pdb     NAMRIILLGAPGAGKGTQAQFIMEKFGIPQISTGDMRLAA
                                *****  ***** *~*****
                                1      .      .      .      40

[Truncated_Name:1] 1AKE_A.pdb      41      .      .      .      80
[Truncated_Name:2] 8BQF_A.pdb      VKSGSELGKQAKDIMDAGKLVDELVIALVKERIAQEDCR
[Truncated_Name:3] 4X8M_A.pdb      VKSGSELGKQAKDIMDAGKLVDELVIALVKERIAQE---
[Truncated_Name:4] 6S36_A.pdb      VKSGSELGKQAKDIMDAGKLVDELVIALVKERIAQEDCR
[Truncated_Name:5] 9R6U_A.pdb      VKSGSELGAQAKDIMDAGKLVDELVIALVKERIAQEDCR
[Truncated_Name:6] 9R71_A.pdb      VKSGSELGKQAKDIMDAGKLVDELVIALVKERIAQEDCR
[Truncated_Name:7] 8Q2B_A.pdb      VKSGSELGKQAKDIMDAGKLVDELVIALVKERIAQEDCR
[Truncated_Name:8] 8RJ9_A.pdb      VKSGSELGKQAKDIMDAGKLVDELVIALVKERIAQEDCR
[Truncated_Name:9] 6RZE_A.pdb      VKSGSELGKQAKDIMDAGKLVDELVIALVKERIAQEDCR
[Truncated_Name:10] 4X8H_A.pdb     VKSGSELGKQAKDIMDAGKLVDELVIALVKERIAQEDCR
[Truncated_Name:11] 3HPR_A.pdb     VKSGSELGKQAKDIMDAGKLVDELVIALVKERIAQEDCR
[Truncated_Name:12] 1E4V_A.pdb     VKSGSELGKQAKDIMDAGKLVDELVIALVKERIAQEDCR
```

[Truncated_Name:13] 5EJE_A.pdb	VKSGSELGKQAKDIMDACKLVTDELVIALVKERIAQEDCR	
[Truncated_Name:14] 1E4Y_A.pdb	VKSGSELGKQAKDIMDAGKLVDELVIALVKERIAQEDCR	
[Truncated_Name:15] 3X2S_A.pdb	VKSGSELGKQAKDIMDCGKLVDELVIALVKERIAQEDSR	
[Truncated_Name:16] 6HAP_A.pdb	VKSGSELGKQAKDIMDAGKLVDELVIALVRERICQEDSR	
[Truncated_Name:17] 6HAM_A.pdb	IKSGSELGKQAKDIMDAGKLVDEIIIALVKERICQEDSR	
[Truncated_Name:18] 8PVW_A.pdb	VKSGSELGKQAKDIMDAGKLVDELVIALVKERIAQEDCR	
[Truncated_Name:19] 4K46_A.pdb	IKAGTELGKQAKSVIDAGQLVSDDIILGLVKERIAQDDCA	
[Truncated_Name:20] 4NP6_A.pdb	IKAGTELGKQAKAVIDAGQLVSDDIILGLIKERIAQADCE	
	~* *~*** ** ~* **~*~*~*~*~*~*~*~*~*	
	41 . . . 80	
	81 . . . 120	
[Truncated_Name:1] 1AKE_A.pdb	NGFLLDGFPR TIPQADAMKEAGINVDYVLEFDVPDELIVD	
[Truncated_Name:2] 8BQF_A.pdb	-GFLLDGFPR TIPQADAMKEAGINVDYVIEFDVPDELIVD	
[Truncated_Name:3] 4X8M_A.pdb	NGFLLDGFPR TIPQADAMKEAGINVDYVLEFDVPDELIVD	
[Truncated_Name:4] 6S36_A.pdb	NGFLLDGFPR TIPQADAMKEAGINVDYVLEFDVPDELIVD	
[Truncated_Name:5] 9R6U_A.pdb	NGFLLDGFPR TIPQADAMKEAGINVDYVLEFDVPDELIVD	
[Truncated_Name:6] 9R71_A.pdb	NGFLLDGFPR TIPQADAMKEAGINVDYVLEFDVPDALIVD	
[Truncated_Name:7] 8Q2B_A.pdb	NGFLLDGFPR TIPQADAMKEAGINVDYVLEFDVPDELIVD	
[Truncated_Name:8] 8RJ9_A.pdb	NGFLLAGFPR TIPQADAMKEAGINVDYVLEFDVPDELIVD	
[Truncated_Name:9] 6RZE_A.pdb	NGFLLDGFPR TIPQADAMKEAGINVDYVLEFDVPDELIVD	
[Truncated_Name:10] 4X8H_A.pdb	NGFLLDGFPR TIPQADAMKEAGINVDYVLEFDVPDELIVD	
[Truncated_Name:11] 3HPR_A.pdb	NGFLLDGFPR TIPQADAMKEAGINVDYVLEFDVPDELIVD	
[Truncated_Name:12] 1E4V_A.pdb	NGFLLDGFPR TIPQADAMKEAGINVDYVLEFDVPDELIVD	
[Truncated_Name:13] 5EJE_A.pdb	NGFLLDGFPR TIPQADAMKEAGINVDYVLEFDVPDELIVD	
[Truncated_Name:14] 1E4Y_A.pdb	NGFLLDGFPR TIPQADAMKEAGINVDYVLEFDVPDELIVD	
[Truncated_Name:15] 3X2S_A.pdb	NGFLLDGFPR TIPQADAMKEAGINVDYVLEFDVPDELIVD	
[Truncated_Name:16] 6HAP_A.pdb	NGFLLDGFPR TIPQADAMKEAGINVDYVLEFDVPDELIVD	
[Truncated_Name:17] 6HAM_A.pdb	NGFLLDGFPR TIPQADAMKEAGINVDYVLEFDVPDELIVD	
[Truncated_Name:18] 8PVW_A.pdb	NGFLLDGFPR TIPQADAMKEAGINVDYVLEFDVPDELIVD	
[Truncated_Name:19] 4K46_A.pdb	KGFLLDGFPR TIPQADGLKEVG VVDYVIEFDVADSVIVE	
[Truncated_Name:20] 4NP6_A.pdb	KGFLLDGFPR TIPQADGLKEMGINVDYVIEFDVADSVIVE	
	**** *****~*~*~*~*~*~*~*~*~*~*~*~*~*~*~*~*	
	81 . . . 120	
	121 . . . 160	
[Truncated_Name:1] 1AKE_A.pdb	RIVGRRVHAPSGRVYHV KFNPPKVEGKDDVTGEELTTRKD	
[Truncated_Name:2] 8BQF_A.pdb	RIVGRRVHAPSGRVYHV KFNPPKVEGKDDVTGEELTTRKD	
[Truncated_Name:3] 4X8M_A.pdb	RIVGRRVHAPSGRVYHV KFNPPKVEGKDDVTGEELTTRKD	
[Truncated_Name:4] 6S36_A.pdb	KIVGRRVHAPSGRVYHV KFNPPKVEGKDDVTGEELTTRKD	
[Truncated_Name:5] 9R6U_A.pdb	RIVGRRVHAPSGRVYHV KFNPPKVEGKDDVTGEELTTRKD	
[Truncated_Name:6] 9R71_A.pdb	RIVGRRVHAPSGRVYHV KFNPPKVEGKDDVTGEELTTRKD	
[Truncated_Name:7] 8Q2B_A.pdb	RIVGRRVHAPSGRVYHV KFNPPKVEGKDDVTGEELTTRKA	

[Truncated_Name:8] 8RJ9_A.pdb	RIVGRRVHAPSGRVYHVKNPPKVEGKDDVTGEELTTRKD
[Truncated_Name:9] 6RZE_A.pdb	AIVGRRVHAPSGRVYHVKNPPKVEGKDDVTGEELTTRKD
[Truncated_Name:10] 4X8H_A.pdb	RIVGRRVHAPSGRVYHVKNPPKVEGKDDVTGEELTTRKD
[Truncated_Name:11] 3HPR_A.pdb	RIVGRRVHAPSGRVYHVKNPPKVEGKDDGTGEELTTRKD
[Truncated_Name:12] 1E4V_A.pdb	RIVGRRVHAPSGRVYHVKNPPKVEGKDDVTGEELTTRKD
[Truncated_Name:13] 5EJE_A.pdb	RIVGRRVHAPSGRVYHVKNPPKVEGKDDVTGEELTTRKD
[Truncated_Name:14] 1E4Y_A.pdb	RIVGRRVHAPSGRVYHVKNPPKVEGKDDVTGEELTTRKD
[Truncated_Name:15] 3X2S_A.pdb	RIVGRRVHAPSGRVYHVKNPPKVEGKDDVTGEELTTRKD
[Truncated_Name:16] 6HAP_A.pdb	RIVGRRVHAPSGRVYHVKNPPKVEGKDDVTGEELTTRKD
[Truncated_Name:17] 6HAM_A.pdb	RIVGRRVHAPSGRVYHVKNPPKVEGKDDVTGEELTTRKD
[Truncated_Name:18] 8PVW_A.pdb	RILKR--GETSGRV-----D
[Truncated_Name:19] 4K46_A.pdb	RMAGRRAHLASGRTYHNVNPPKVEGKDDVTGEDLVIRED
[Truncated_Name:20] 4NP6_A.pdb	RMAGRRAHLPSGRTYHVVNPPKVEGKDDVTGEDLVIRED
	^ * ***
	121 . . . 160
	161 . . . 200
[Truncated_Name:1] 1AKE_A.pdb	DQEETVRKRLVEYHQM TAPLIGYYSKEAEAGNTKYAKVDG
[Truncated_Name:2] 8BQF_A.pdb	DQEETVRKRLVEYHQM TAPLIGYYSKEAEAGNTKYAKVDG
[Truncated_Name:3] 4X8M_A.pdb	DQEETVRKRLVEWHQM TAPLIGYYSKEAEAGNTKYAKVDG
[Truncated_Name:4] 6S36_A.pdb	DQEETVRKRLVEYHQM TAPLIGYYSKEAEAGNTKYAKVDG
[Truncated_Name:5] 9R6U_A.pdb	DQEETVRKRLVEYHQM TAPLIGYYSKEAEAGNTKYAKVDG
[Truncated_Name:6] 9R71_A.pdb	DQEETVRKRLVEYHQM TAPLIGYYSKEAEAGNTKYAKVDG
[Truncated_Name:7] 8Q2B_A.pdb	DQEETVRKRLVEYHQM TAPLIGYYSKEAEAGNTKYAKVDG
[Truncated_Name:8] 8RJ9_A.pdb	DQEETVRKRLVEYHQM TAPLIGYYSKEAEAGNTKYAKVDG
[Truncated_Name:9] 6RZE_A.pdb	DQEETVRKRLVEYHQM TAPLIGYYSKEAEAGNTKYAKVDG
[Truncated_Name:10] 4X8H_A.pdb	DQEETVRKRLVEYHQM TAPLIGYYSKEAEAGNTKYAKVDG
[Truncated_Name:11] 3HPR_A.pdb	DQEETVRKRLVEYHQM TAPLIGYYSKEAEAGNTKYAKVDG
[Truncated_Name:12] 1E4V_A.pdb	DQEETVRKRLVEYHQM TAPLIGYYSKEAEAGNTKYAKVDG
[Truncated_Name:13] 5EJE_A.pdb	DQEECVRKRLVEYHQM TAPLIGYYSKEAEAGNTKYAKVDG
[Truncated_Name:14] 1E4Y_A.pdb	DQEETVRKRLVEYHQM TAPLIGYYSKEAEAGNTKYAKVDG
[Truncated_Name:15] 3X2S_A.pdb	DQEETVRKRLCEYHQM TAPLIGYYSKEAEAGNTKYAKVDG
[Truncated_Name:16] 6HAP_A.pdb	DQEETVRKRLVEYHQM TAPLIGYYSKEAEAGNTKYAKVDG
[Truncated_Name:17] 6HAM_A.pdb	DQEETVRKRLVEYHQM TAPLIGYYSKEAEAGNTKYAKVDG
[Truncated_Name:18] 8PVW_A.pdb	DNEETVRKRLVEYHQM TAPLIGYYSKEAEAGNTKYAKVDG
[Truncated_Name:19] 4K46_A.pdb	DKEETVLARLGVYHNQTAPLIAYYGKEAEAGNTQYLKFDG
[Truncated_Name:20] 4NP6_A.pdb	DKEETVRARLNVYHTQTAPLIEYYGKEAAAGKTQYLKFDG
	* * * * * ^ * * * * * * * * * *
	161 . . . 200
	201 . 216
[Truncated_Name:1] 1AKE_A.pdb	TKPVAEVRADLEKILG
[Truncated_Name:2] 8BQF_A.pdb	TKPVAEVRADLEKIL-

```

[Truncated_Name:3]4X8M_A.pdb      TKPVAEVRADLEKILG
[Truncated_Name:4]6S36_A.pdb      TKPVAEVRADLEKILG
[Truncated_Name:5]9R6U_A.pdb      TKPVAEVRADLEKILG
[Truncated_Name:6]9R71_A.pdb      TKPVAEVRADLEKILG
[Truncated_Name:7]8Q2B_A.pdb      TKPVAEVRADLEKILG
[Truncated_Name:8]8RJ9_A.pdb      TKPVAEVRADLEKILG
[Truncated_Name:9]6RZE_A.pdb      TKPVAEVRADLEKILG
[Truncated_Name:10]4X8H_A.pdb     TKPVAEVRADLEKILG
[Truncated_Name:11]3HPR_A.pdb     TKPVAEVRADLEKILG
[Truncated_Name:12]1E4V_A.pdb     TKPVAEVRADLEKILG
[Truncated_Name:13]5EJE_A.pdb     TKPVAEVRADLEKILG
[Truncated_Name:14]1E4Y_A.pdb     TKPVAEVRADLEKILG
[Truncated_Name:15]3X2S_A.pdb     TKPVAEVRADLEKILG
[Truncated_Name:16]6HAP_A.pdb     TKPVCEVRADLEKILG
[Truncated_Name:17]6HAM_A.pdb     TKPVCEVRADLEKILG
[Truncated_Name:18]8PVW_A.pdb     TKPVAEVRADLEKILG
[Truncated_Name:19]4K46_A.pdb     TKAVAEVSAELEKALA
[Truncated_Name:20]4NP6_A.pdb     TKQVSEVSADIAKALA
                                ** * ** *~* *
                                201      .      216

```

Call:

```
pdbaln(files = files, fit = TRUE, exefile = "msa")
```

Class:

```
pdb, fasta
```

Alignment dimensions:

```
20 sequence rows; 216 position columns (182 non-gap, 34 gap)
```

```
+ attr: xyz, resno, b, chain, id, ali, resid, sse, call
```

Quick interactive structural view

```
view.pdb(pdb)
```

Warning in basename(x): expanded path length 17340 would be too long for

ATOM	1	CA	MET A	1	26.091	52.849	39.889	1.00	0.00
ATOM	2	CA	ARG A	2	27.437	49.969	37.786	1.00	0.00
ATOM	3	CA	ILE A	3	24.961	47.988	35.671	1.00	0.00
ATOM	4	CA	ILE A	4	25.194	44.925	33.360	1.00	0.00
ATOM	5	CA	LEU A	5	22.428	44.503	30.712	1.00	0.00

ATOM	6	CA	LEU	A	6	21.933	40.811	29.752	1.00	0.00
ATOM	7	CA	GLY	A	7	19.604	39.512	26.973	1.00	0.00
ATOM	8	CA	ALA	A	8	19.324	37.742	23.567	1.00	0.00
ATOM	9	CA	PRO	A	9	20.108	39.651	20.294	1.00	0.00
ATOM	10	CA	GLY	A	10	17.487	42.426	19.756	1.00	0.00
ATOM	11	CA	ALA	A	11	15.960	42.201	23.284	1.00	0.00
ATOM	12	CA	GLY	A	12	16.082	46.035	23.547	1.00	[... truncated]

Warning in basename(x): expanded path length 16935 would be too long for

ATOM	1	CA	MET	A	1	26.009	52.803	39.842	1.00	0.00
ATOM	2	CA	ARG	A	2	27.443	49.839	37.833	1.00	0.00
ATOM	3	CA	ILE	A	3	24.894	47.919	35.662	1.00	0.00
ATOM	4	CA	ILE	A	4	25.078	44.872	33.348	1.00	0.00
ATOM	5	CA	LEU	A	5	22.244	44.445	30.783	1.00	0.00
ATOM	6	CA	LEU	A	6	21.808	40.753	29.741	1.00	0.00
ATOM	7	CA	GLY	A	7	19.605	39.419	26.928	1.00	0.00
ATOM	8	CA	ALA	A	8	19.300	37.881	23.478	1.00	0.00
ATOM	9	CA	PRO	A	9	20.220	39.792	20.316	1.00	0.00
ATOM	10	CA	GLY	A	10	17.490	42.420	19.747	1.00	0.00
ATOM	11	CA	ALA	A	11	16.070	42.223	23.295	1.00	0.00
ATOM	12	CA	GLY	A	12	16.185	46.086	23.537	1.00	[... truncated]

Warning in basename(x): expanded path length 16935 would be too long for

ATOM	1	CA	MET	A	1	26.009	52.803	39.842	1.00	0.00
ATOM	2	CA	ARG	A	2	27.443	49.839	37.833	1.00	0.00
ATOM	3	CA	ILE	A	3	24.894	47.919	35.662	1.00	0.00
ATOM	4	CA	ILE	A	4	25.078	44.872	33.348	1.00	0.00
ATOM	5	CA	LEU	A	5	22.244	44.445	30.783	1.00	0.00
ATOM	6	CA	LEU	A	6	21.808	40.753	29.741	1.00	0.00
ATOM	7	CA	GLY	A	7	19.605	39.419	26.928	1.00	0.00
ATOM	8	CA	ALA	A	8	19.300	37.881	23.478	1.00	0.00
ATOM	9	CA	PRO	A	9	20.220	39.792	20.316	1.00	0.00
ATOM	10	CA	GLY	A	10	17.490	42.420	19.747	1.00	0.00
ATOM	11	CA	ALA	A	11	16.070	42.223	23.295	1.00	0.00
ATOM	12	CA	GLY	A	12	16.185	46.086	23.537	1.00	[... truncated]

Warning in basename(x): expanded path length 17340 would be too long for

ATOM	1	CA	MET	A	1	26.764	52.391	39.761	1.00	0.00
ATOM	2	CA	ARG	A	2	28.147	49.693	37.465	1.00	0.00
ATOM	3	CA	ILE	A	3	25.482	47.950	35.382	1.00	0.00
ATOM	4	CA	ILE	A	4	25.524	45.048	32.936	1.00	0.00
ATOM	5	CA	LEU	A	5	22.727	45.104	30.357	1.00	0.00
ATOM	6	CA	LEU	A	6	21.909	41.757	28.733	1.00	0.00

ATOM	7	CA	GLY	A	7	19.146	40.959	26.250	1.00	0.00
ATOM	8	CA	ALA	A	8	18.220	39.073	23.087	1.00	0.00
ATOM	9	CA	PRO	A	9	18.647	40.652	19.634	1.00	0.00
ATOM	10	CA	GLY	A	10	15.577	42.833	19.065	1.00	0.00
ATOM	11	CA	ALA	A	11	14.520	42.767	22.717	1.00	0.00
ATOM	12	CA	GLY	A	12	15.075	46.528	22.818	1.00	[... truncated]

Warning in basename(x): expanded path length 17340 would be too long for

ATOM	1	CA	MET	A	1	26.764	52.391	39.761	1.00	0.00
ATOM	2	CA	ARG	A	2	28.147	49.693	37.465	1.00	0.00
ATOM	3	CA	ILE	A	3	25.482	47.950	35.382	1.00	0.00
ATOM	4	CA	ILE	A	4	25.524	45.048	32.936	1.00	0.00
ATOM	5	CA	LEU	A	5	22.727	45.104	30.357	1.00	0.00
ATOM	6	CA	LEU	A	6	21.909	41.757	28.733	1.00	0.00
ATOM	7	CA	GLY	A	7	19.146	40.959	26.250	1.00	0.00
ATOM	8	CA	ALA	A	8	18.220	39.073	23.087	1.00	0.00
ATOM	9	CA	PRO	A	9	18.647	40.652	19.634	1.00	0.00
ATOM	10	CA	GLY	A	10	15.577	42.833	19.065	1.00	0.00
ATOM	11	CA	ALA	A	11	14.520	42.767	22.717	1.00	0.00
ATOM	12	CA	GLY	A	12	15.075	46.528	22.818	1.00	[... truncated]

Warning in basename(x): expanded path length 17340 would be too long for

ATOM	1	CA	MET	A	1	26.594	52.601	39.133	1.00	0.00
ATOM	2	CA	ARG	A	2	28.189	49.782	37.141	1.00	0.00
ATOM	3	CA	ILE	A	3	25.532	47.995	35.098	1.00	0.00
ATOM	4	CA	ILE	A	4	25.550	45.126	32.622	1.00	0.00
ATOM	5	CA	LEU	A	5	22.548	45.208	30.263	1.00	0.00
ATOM	6	CA	LEU	A	6	21.836	41.950	28.421	1.00	0.00
ATOM	7	CA	GLY	A	7	19.146	41.165	25.906	1.00	0.00
ATOM	8	CA	ALA	A	8	18.345	39.054	22.895	1.00	0.00
ATOM	9	CA	PRO	A	9	18.870	40.516	19.404	1.00	0.00
ATOM	10	CA	GLY	A	10	15.714	42.429	18.527	1.00	0.00
ATOM	11	CA	ALA	A	11	14.290	42.679	22.043	1.00	0.00
ATOM	12	CA	GLY	A	12	14.976	46.431	22.244	1.00	[... truncated]

Warning in basename(x): expanded path length 17340 would be too long for

ATOM	1	CA	MET	A	1	26.594	52.601	39.133	1.00	0.00
ATOM	2	CA	ARG	A	2	28.189	49.782	37.141	1.00	0.00
ATOM	3	CA	ILE	A	3	25.532	47.995	35.098	1.00	0.00
ATOM	4	CA	ILE	A	4	25.550	45.126	32.622	1.00	0.00
ATOM	5	CA	LEU	A	5	22.548	45.208	30.263	1.00	0.00
ATOM	6	CA	LEU	A	6	21.836	41.950	28.421	1.00	0.00
ATOM	7	CA	GLY	A	7	19.146	41.165	25.906	1.00	0.00
ATOM	8	CA	ALA	A	8	18.345	39.054	22.895	1.00	0.00

ATOM	9	CA	PRO	A	9	18.870	40.516	19.404	1.00	0.00
ATOM	10	CA	GLY	A	10	15.714	42.429	18.527	1.00	0.00
ATOM	11	CA	ALA	A	11	14.290	42.679	22.043	1.00	0.00
ATOM	12	CA	GLY	A	12	14.976	46.431	22.244	1.00	[... truncated]

Warning in basename(x): expanded path length 17340 would be too long for

ATOM	1	CA	MET	A	1	25.885	53.058	40.012	1.00	0.00
ATOM	2	CA	ARG	A	2	27.197	50.121	37.922	1.00	0.00
ATOM	3	CA	ILE	A	3	24.804	48.064	35.763	1.00	0.00
ATOM	4	CA	ILE	A	4	25.020	45.026	33.436	1.00	0.00
ATOM	5	CA	LEU	A	5	22.331	44.369	30.796	1.00	0.00
ATOM	6	CA	LEU	A	6	21.693	40.725	29.793	1.00	0.00
ATOM	7	CA	GLY	A	7	19.504	39.278	27.030	1.00	0.00
ATOM	8	CA	ALA	A	8	19.314	37.674	23.600	1.00	0.00
ATOM	9	CA	PRO	A	9	20.060	39.544	20.342	1.00	0.00
ATOM	10	CA	GLY	A	10	17.363	42.127	19.647	1.00	0.00
ATOM	11	CA	ALA	A	11	15.915	42.058	23.183	1.00	0.00
ATOM	12	CA	GLY	A	12	15.987	45.883	23.500	1.00	[... truncated]

Warning in basename(x): expanded path length 17340 would be too long for

ATOM	1	CA	MET	A	1	25.885	53.058	40.012	1.00	0.00
ATOM	2	CA	ARG	A	2	27.197	50.121	37.922	1.00	0.00
ATOM	3	CA	ILE	A	3	24.804	48.064	35.763	1.00	0.00
ATOM	4	CA	ILE	A	4	25.020	45.026	33.436	1.00	0.00
ATOM	5	CA	LEU	A	5	22.331	44.369	30.796	1.00	0.00
ATOM	6	CA	LEU	A	6	21.693	40.725	29.793	1.00	0.00
ATOM	7	CA	GLY	A	7	19.504	39.278	27.030	1.00	0.00
ATOM	8	CA	ALA	A	8	19.314	37.674	23.600	1.00	0.00
ATOM	9	CA	PRO	A	9	20.060	39.544	20.342	1.00	0.00
ATOM	10	CA	GLY	A	10	17.363	42.127	19.647	1.00	0.00
ATOM	11	CA	ALA	A	11	15.915	42.058	23.183	1.00	0.00
ATOM	12	CA	GLY	A	12	15.987	45.883	23.500	1.00	[... truncated]

Warning in basename(x): expanded path length 17340 would be too long for

ATOM	1	CA	MET	A	1	26.336	53.102	39.673	1.00	0.00
ATOM	2	CA	ARG	A	2	27.453	50.072	37.643	1.00	0.00
ATOM	3	CA	ILE	A	3	24.920	48.062	35.654	1.00	0.00
ATOM	4	CA	ILE	A	4	25.064	45.066	33.307	1.00	0.00
ATOM	5	CA	LEU	A	5	22.383	44.572	30.650	1.00	0.00
ATOM	6	CA	LEU	A	6	21.771	40.926	29.710	1.00	0.00
ATOM	7	CA	GLY	A	7	19.604	39.525	26.953	1.00	0.00
ATOM	8	CA	ALA	A	8	19.363	37.837	23.593	1.00	0.00
ATOM	9	CA	PRO	A	9	20.121	39.758	20.399	1.00	0.00

ATOM	10	CA	GLY	A	10	17.412	42.298	19.707	1.00	0.00
ATOM	11	CA	ALA	A	11	15.974	42.151	23.223	1.00	0.00
ATOM	12	CA	GLY	A	12	16.064	45.953	23.598	1.00	[... truncated]

Warning in basename(x): expanded path length 17340 would be too long for

ATOM	1	CA	MET	A	1	26.336	53.102	39.673	1.00	0.00
ATOM	2	CA	ARG	A	2	27.453	50.072	37.643	1.00	0.00
ATOM	3	CA	ILE	A	3	24.920	48.062	35.654	1.00	0.00
ATOM	4	CA	ILE	A	4	25.064	45.066	33.307	1.00	0.00
ATOM	5	CA	LEU	A	5	22.383	44.572	30.650	1.00	0.00
ATOM	6	CA	LEU	A	6	21.771	40.926	29.710	1.00	0.00
ATOM	7	CA	GLY	A	7	19.604	39.525	26.953	1.00	0.00
ATOM	8	CA	ALA	A	8	19.363	37.837	23.593	1.00	0.00
ATOM	9	CA	PRO	A	9	20.121	39.758	20.399	1.00	0.00
ATOM	10	CA	GLY	A	10	17.412	42.298	19.707	1.00	0.00
ATOM	11	CA	ALA	A	11	15.974	42.151	23.223	1.00	0.00
ATOM	12	CA	GLY	A	12	16.064	45.953	23.598	1.00	[... truncated]

Warning in basename(x): expanded path length 17340 would be too long for

ATOM	1	CA	MET	A	1	26.292	53.130	40.000	1.00	0.00
ATOM	2	CA	ARG	A	2	27.458	50.072	37.962	1.00	0.00
ATOM	3	CA	ILE	A	3	24.878	48.118	35.897	1.00	0.00
ATOM	4	CA	ILE	A	4	25.070	45.076	33.547	1.00	0.00
ATOM	5	CA	LEU	A	5	22.349	44.642	30.864	1.00	0.00
ATOM	6	CA	LEU	A	6	21.858	40.943	29.929	1.00	0.00
ATOM	7	CA	GLY	A	7	19.767	39.676	26.995	1.00	0.00
ATOM	8	CA	ALA	A	8	19.588	37.993	23.569	1.00	0.00
ATOM	9	CA	PRO	A	9	21.101	39.899	20.601	1.00	0.00
ATOM	10	CA	GLY	A	10	18.115	42.050	19.416	1.00	0.00
ATOM	11	CA	ALA	A	11	16.162	42.059	22.763	1.00	0.00
ATOM	12	CA	GLY	A	12	16.197	45.928	22.965	1.00	[... truncated]

Warning in basename(x): expanded path length 17340 would be too long for

ATOM	1	CA	MET	A	1	26.292	53.130	40.000	1.00	0.00
ATOM	2	CA	ARG	A	2	27.458	50.072	37.962	1.00	0.00
ATOM	3	CA	ILE	A	3	24.878	48.118	35.897	1.00	0.00
ATOM	4	CA	ILE	A	4	25.070	45.076	33.547	1.00	0.00
ATOM	5	CA	LEU	A	5	22.349	44.642	30.864	1.00	0.00
ATOM	6	CA	LEU	A	6	21.858	40.943	29.929	1.00	0.00
ATOM	7	CA	GLY	A	7	19.767	39.676	26.995	1.00	0.00
ATOM	8	CA	ALA	A	8	19.588	37.993	23.569	1.00	0.00
ATOM	9	CA	PRO	A	9	21.101	39.899	20.601	1.00	0.00
ATOM	10	CA	GLY	A	10	18.115	42.050	19.416	1.00	0.00
ATOM	11	CA	ALA	A	11	16.162	42.059	22.763	1.00	0.00
ATOM	12	CA	GLY	A	12	16.197	45.928	22.965	1.00	[... truncated]

Warning in basename(x): expanded path length 17340 would be too long for

ATOM	1	CA	MET	A	1	25.950	52.850	39.782	1.00	0.00
ATOM	2	CA	ARG	A	2	27.196	49.918	37.712	1.00	0.00
ATOM	3	CA	ILE	A	3	24.772	47.896	35.575	1.00	0.00
ATOM	4	CA	ILE	A	4	24.910	44.845	33.313	1.00	0.00
ATOM	5	CA	LEU	A	5	22.088	44.317	30.820	1.00	0.00
ATOM	6	CA	LEU	A	6	21.611	40.656	29.864	1.00	0.00
ATOM	7	CA	GLY	A	7	19.496	39.249	27.057	1.00	0.00
ATOM	8	CA	ALA	A	8	19.457	37.795	23.562	1.00	0.00
ATOM	9	CA	PRO	A	9	20.408	39.856	20.530	1.00	0.00
ATOM	10	CA	GLY	A	10	17.479	42.110	19.809	1.00	0.00
ATOM	11	CA	ALA	A	11	16.078	41.995	23.352	1.00	0.00
ATOM	12	CA	GLY	A	12	16.228	45.821	23.710	1.00	[... truncated]

Warning in basename(x): expanded path length 17340 would be too long for

ATOM	1	CA	MET	A	1	25.950	52.850	39.782	1.00	0.00
ATOM	2	CA	ARG	A	2	27.196	49.918	37.712	1.00	0.00
ATOM	3	CA	ILE	A	3	24.772	47.896	35.575	1.00	0.00
ATOM	4	CA	ILE	A	4	24.910	44.845	33.313	1.00	0.00
ATOM	5	CA	LEU	A	5	22.088	44.317	30.820	1.00	0.00
ATOM	6	CA	LEU	A	6	21.611	40.656	29.864	1.00	0.00
ATOM	7	CA	GLY	A	7	19.496	39.249	27.057	1.00	0.00
ATOM	8	CA	ALA	A	8	19.457	37.795	23.562	1.00	0.00
ATOM	9	CA	PRO	A	9	20.408	39.856	20.530	1.00	0.00
ATOM	10	CA	GLY	A	10	17.479	42.110	19.809	1.00	0.00
ATOM	11	CA	ALA	A	11	16.078	41.995	23.352	1.00	0.00
ATOM	12	CA	GLY	A	12	16.228	45.821	23.710	1.00	[... truncated]

Warning in basename(x): expanded path length 17340 would be too long for

ATOM	1	CA	MET	A	1	26.926	52.755	39.454	1.00	0.00
ATOM	2	CA	ARG	A	2	28.148	49.780	37.355	1.00	0.00
ATOM	3	CA	ILE	A	3	25.518	48.030	35.221	1.00	0.00
ATOM	4	CA	ILE	A	4	25.577	45.165	32.745	1.00	0.00
ATOM	5	CA	LEU	A	5	22.609	45.338	30.338	1.00	0.00
ATOM	6	CA	LEU	A	6	21.846	42.094	28.489	1.00	0.00
ATOM	7	CA	GLY	A	7	19.207	41.270	25.919	1.00	0.00
ATOM	8	CA	ALA	A	8	18.348	39.244	22.822	1.00	0.00
ATOM	9	CA	PRO	A	9	18.962	40.723	19.359	1.00	0.00
ATOM	10	CA	GLY	A	10	15.807	42.614	18.444	1.00	0.00
ATOM	11	CA	ALA	A	11	14.426	42.776	21.991	1.00	0.00
ATOM	12	CA	GLY	A	12	15.103	46.511	22.136	1.00	[... truncated]

Warning in basename(x): expanded path length 17340 would be too long for

ATOM	1	CA	MET	A	1	26.926	52.755	39.454	1.00	0.00
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ATOM	2	CA	ARG	A	2	28.148	49.780	37.355	1.00	0.00
ATOM	3	CA	ILE	A	3	25.518	48.030	35.221	1.00	0.00
ATOM	4	CA	ILE	A	4	25.577	45.165	32.745	1.00	0.00
ATOM	5	CA	LEU	A	5	22.609	45.338	30.338	1.00	0.00
ATOM	6	CA	LEU	A	6	21.846	42.094	28.489	1.00	0.00
ATOM	7	CA	GLY	A	7	19.207	41.270	25.919	1.00	0.00
ATOM	8	CA	ALA	A	8	18.348	39.244	22.822	1.00	0.00
ATOM	9	CA	PRO	A	9	18.962	40.723	19.359	1.00	0.00
ATOM	10	CA	GLY	A	10	15.807	42.614	18.444	1.00	0.00
ATOM	11	CA	ALA	A	11	14.426	42.776	21.991	1.00	0.00
ATOM	12	CA	GLY	A	12	15.103	46.511	22.136	1.00	[... truncated]

Warning in basename(x): expanded path length 17340 would be too long for

ATOM	1	CA	MET	A	1	27.070	52.649	39.229	1.00	0.00
ATOM	2	CA	ARG	A	2	28.244	49.743	37.067	1.00	0.00
ATOM	3	CA	ILE	A	3	25.555	47.979	35.032	1.00	0.00
ATOM	4	CA	ILE	A	4	25.518	45.050	32.615	1.00	0.00
ATOM	5	CA	LEU	A	5	22.569	45.136	30.212	1.00	0.00
ATOM	6	CA	LEU	A	6	21.702	41.852	28.489	1.00	0.00
ATOM	7	CA	GLY	A	7	18.950	40.946	26.043	1.00	0.00
ATOM	8	CA	ALA	A	8	18.120	38.888	22.965	1.00	0.00
ATOM	9	CA	PRO	A	9	18.785	40.376	19.494	1.00	0.00
ATOM	10	CA	GLY	A	10	15.736	42.439	18.538	1.00	0.00
ATOM	11	CA	ALA	A	11	14.298	42.601	22.046	1.00	0.00
ATOM	12	CA	GLY	A	12	15.058	46.331	22.150	1.00	[... truncated]

Warning in basename(x): expanded path length 17340 would be too long for

ATOM	1	CA	MET	A	1	27.070	52.649	39.229	1.00	0.00
ATOM	2	CA	ARG	A	2	28.244	49.743	37.067	1.00	0.00
ATOM	3	CA	ILE	A	3	25.555	47.979	35.032	1.00	0.00
ATOM	4	CA	ILE	A	4	25.518	45.050	32.615	1.00	0.00
ATOM	5	CA	LEU	A	5	22.569	45.136	30.212	1.00	0.00
ATOM	6	CA	LEU	A	6	21.702	41.852	28.489	1.00	0.00
ATOM	7	CA	GLY	A	7	18.950	40.946	26.043	1.00	0.00
ATOM	8	CA	ALA	A	8	18.120	38.888	22.965	1.00	0.00
ATOM	9	CA	PRO	A	9	18.785	40.376	19.494	1.00	0.00
ATOM	10	CA	GLY	A	10	15.736	42.439	18.538	1.00	0.00
ATOM	11	CA	ALA	A	11	14.298	42.601	22.046	1.00	0.00
ATOM	12	CA	GLY	A	12	15.058	46.331	22.150	1.00	[... truncated]

Warning in basename(x): expanded path length 17340 would be too long for

ATOM	1	CA	MET	A	1	26.434	52.839	39.971	1.00	0.00
ATOM	2	CA	ARG	A	2	27.414	49.883	37.798	1.00	0.00

ATOM	3	CA	ILE	A	3	24.943	47.903	35.698	1.00	0.00
ATOM	4	CA	ILE	A	4	25.140	44.959	33.307	1.00	0.00
ATOM	5	CA	LEU	A	5	22.459	44.482	30.644	1.00	0.00
ATOM	6	CA	LEU	A	6	21.791	40.845	29.749	1.00	0.00
ATOM	7	CA	GLY	A	7	19.635	39.255	27.071	1.00	0.00
ATOM	8	CA	ALA	A	8	19.204	37.793	23.601	1.00	0.00
ATOM	9	CA	PRO	A	9	19.988	39.600	20.324	1.00	0.00
ATOM	10	CA	GLY	A	10	17.411	42.332	19.730	1.00	0.00
ATOM	11	CA	ALA	A	11	16.015	42.075	23.258	1.00	0.00
ATOM	12	CA	GLY	A	12	16.144	45.876	23.640	1.00	[... truncated]

Warning in basename(x): expanded path length 17340 would be too long for

ATOM	1	CA	MET	A	1	26.434	52.839	39.971	1.00	0.00
ATOM	2	CA	ARG	A	2	27.414	49.883	37.798	1.00	0.00
ATOM	3	CA	ILE	A	3	24.943	47.903	35.698	1.00	0.00
ATOM	4	CA	ILE	A	4	25.140	44.959	33.307	1.00	0.00
ATOM	5	CA	LEU	A	5	22.459	44.482	30.644	1.00	0.00
ATOM	6	CA	LEU	A	6	21.791	40.845	29.749	1.00	0.00
ATOM	7	CA	GLY	A	7	19.635	39.255	27.071	1.00	0.00
ATOM	8	CA	ALA	A	8	19.204	37.793	23.601	1.00	0.00
ATOM	9	CA	PRO	A	9	19.988	39.600	20.324	1.00	0.00
ATOM	10	CA	GLY	A	10	17.411	42.332	19.730	1.00	0.00
ATOM	11	CA	ALA	A	11	16.015	42.075	23.258	1.00	0.00
ATOM	12	CA	GLY	A	12	16.144	45.876	23.640	1.00	[... truncated]

Warning in basename(x): expanded path length 17340 would be too long for

ATOM	1	CA	MET	A	1	26.169	52.665	39.983	1.00	0.00
ATOM	2	CA	ARG	A	2	27.421	49.874	37.676	1.00	0.00
ATOM	3	CA	ILE	A	3	24.867	47.934	35.638	1.00	0.00
ATOM	4	CA	ILE	A	4	25.228	44.932	33.277	1.00	0.00
ATOM	5	CA	LEU	A	5	22.456	44.545	30.631	1.00	0.00
ATOM	6	CA	LEU	A	6	22.157	40.853	29.716	1.00	0.00
ATOM	7	CA	GLY	A	7	19.842	39.396	26.994	1.00	0.00
ATOM	8	CA	ALA	A	8	19.267	37.785	23.513	1.00	0.00
ATOM	9	CA	PRO	A	9	20.049	39.447	20.186	1.00	0.00
ATOM	10	CA	VAL	A	10	17.828	42.618	19.709	1.00	0.00
ATOM	11	CA	ALA	A	11	16.154	42.333	23.069	1.00	0.00
ATOM	12	CA	GLY	A	12	16.208	46.140	23.338	1.00	[... truncated]

Warning in basename(x): expanded path length 17340 would be too long for

ATOM	1	CA	MET	A	1	26.169	52.665	39.983	1.00	0.00
ATOM	2	CA	ARG	A	2	27.421	49.874	37.676	1.00	0.00
ATOM	3	CA	ILE	A	3	24.867	47.934	35.638	1.00	0.00
ATOM	4	CA	ILE	A	4	25.228	44.932	33.277	1.00	0.00

ATOM	5	CA	LEU	A	5	22.456	44.545	30.631	1.00	0.00
ATOM	6	CA	LEU	A	6	22.157	40.853	29.716	1.00	0.00
ATOM	7	CA	GLY	A	7	19.842	39.396	26.994	1.00	0.00
ATOM	8	CA	ALA	A	8	19.267	37.785	23.513	1.00	0.00
ATOM	9	CA	PRO	A	9	20.049	39.447	20.186	1.00	0.00
ATOM	10	CA	VAL	A	10	17.828	42.618	19.709	1.00	0.00
ATOM	11	CA	ALA	A	11	16.154	42.333	23.069	1.00	0.00
ATOM	12	CA	GLY	A	12	16.208	46.140	23.338	1.00	[... truncated]

Warning in basename(x): expanded path length 17340 would be too long for

ATOM	1	CA	MET	A	1	26.206	52.988	39.647	1.00	0.00
ATOM	2	CA	ARG	A	2	27.415	49.971	37.651	1.00	0.00
ATOM	3	CA	ILE	A	3	24.919	47.976	35.573	1.00	0.00
ATOM	4	CA	ILE	A	4	25.060	44.902	33.343	1.00	0.00
ATOM	5	CA	LEU	A	5	22.315	44.372	30.765	1.00	0.00
ATOM	6	CA	LEU	A	6	21.767	40.707	29.832	1.00	0.00
ATOM	7	CA	GLY	A	7	19.630	39.546	26.936	1.00	0.00
ATOM	8	CA	ALA	A	8	19.508	37.898	23.556	1.00	0.00
ATOM	9	CA	PRO	A	9	20.500	39.786	20.407	1.00	0.00
ATOM	10	CA	GLY	A	10	17.712	42.240	19.682	1.00	0.00
ATOM	11	CA	ALA	A	11	16.205	42.131	23.183	1.00	0.00
ATOM	12	CA	GLY	A	12	16.410	45.927	23.471	1.00	[... truncated]

Warning in basename(x): expanded path length 17340 would be too long for

ATOM	1	CA	MET	A	1	26.206	52.988	39.647	1.00	0.00
ATOM	2	CA	ARG	A	2	27.415	49.971	37.651	1.00	0.00
ATOM	3	CA	ILE	A	3	24.919	47.976	35.573	1.00	0.00
ATOM	4	CA	ILE	A	4	25.060	44.902	33.343	1.00	0.00
ATOM	5	CA	LEU	A	5	22.315	44.372	30.765	1.00	0.00
ATOM	6	CA	LEU	A	6	21.767	40.707	29.832	1.00	0.00
ATOM	7	CA	GLY	A	7	19.630	39.546	26.936	1.00	0.00
ATOM	8	CA	ALA	A	8	19.508	37.898	23.556	1.00	0.00
ATOM	9	CA	PRO	A	9	20.500	39.786	20.407	1.00	0.00
ATOM	10	CA	GLY	A	10	17.712	42.240	19.682	1.00	0.00
ATOM	11	CA	ALA	A	11	16.205	42.131	23.183	1.00	0.00
ATOM	12	CA	GLY	A	12	16.410	45.927	23.471	1.00	[... truncated]

Warning in basename(x): expanded path length 17340 would be too long for

ATOM	1	CA	MET	A	1	26.470	53.073	40.047	1.00	0.00
ATOM	2	CA	ARG	A	2	27.372	49.946	37.943	1.00	0.00
ATOM	3	CA	ILE	A	3	24.805	47.939	35.973	1.00	0.00
ATOM	4	CA	ILE	A	4	25.065	45.020	33.522	1.00	0.00
ATOM	5	CA	LEU	A	5	22.295	44.779	30.878	1.00	0.00

ATOM	6	CA	LEU	A	6	22.024	41.154	29.865	1.00	0.00
ATOM	7	CA	GLY	A	7	19.348	39.827	27.537	1.00	0.00
ATOM	8	CA	ALA	A	8	18.423	37.934	24.441	1.00	0.00
ATOM	9	CA	LEU	A	9	19.199	39.236	21.021	1.00	0.00
ATOM	10	CA	VAL	A	10	17.112	42.248	20.017	1.00	0.00
ATOM	11	CA	ALA	A	11	15.347	42.365	23.436	1.00	0.00
ATOM	12	CA	GLY	A	12	15.834	46.161	23.646	1.00	[... truncated]

Warning in basename(x): expanded path length 17340 would be too long for

ATOM	1	CA	MET	A	1	26.470	53.073	40.047	1.00	0.00
ATOM	2	CA	ARG	A	2	27.372	49.946	37.943	1.00	0.00
ATOM	3	CA	ILE	A	3	24.805	47.939	35.973	1.00	0.00
ATOM	4	CA	ILE	A	4	25.065	45.020	33.522	1.00	0.00
ATOM	5	CA	LEU	A	5	22.295	44.779	30.878	1.00	0.00
ATOM	6	CA	LEU	A	6	22.024	41.154	29.865	1.00	0.00
ATOM	7	CA	GLY	A	7	19.348	39.827	27.537	1.00	0.00
ATOM	8	CA	ALA	A	8	18.423	37.934	24.441	1.00	0.00
ATOM	9	CA	LEU	A	9	19.199	39.236	21.021	1.00	0.00
ATOM	10	CA	VAL	A	10	17.112	42.248	20.017	1.00	0.00
ATOM	11	CA	ALA	A	11	15.347	42.365	23.436	1.00	0.00
ATOM	12	CA	GLY	A	12	15.834	46.161	23.646	1.00	[... truncated]

Warning in basename(x): expanded path length 17340 would be too long for

ATOM	1	CA	MET	A	1	26.318	52.948	39.518	1.00	0.00
ATOM	2	CA	ARG	A	2	27.607	49.903	37.679	1.00	0.00
ATOM	3	CA	ILE	A	3	25.007	47.987	35.714	1.00	0.00
ATOM	4	CA	ILE	A	4	25.025	44.972	33.400	1.00	0.00
ATOM	5	CA	LEU	A	5	22.465	44.494	30.650	1.00	0.00
ATOM	6	CA	LEU	A	6	22.143	40.787	29.919	1.00	0.00
ATOM	7	CA	GLY	A	7	20.072	39.724	26.945	1.00	0.00
ATOM	8	CA	ALA	A	8	19.966	38.397	23.396	1.00	0.00
ATOM	9	CA	PRO	A	9	20.372	40.256	20.073	1.00	0.00
ATOM	10	CA	GLY	A	10	17.506	42.659	19.351	1.00	0.00
ATOM	11	CA	ALA	A	11	16.395	42.374	23.007	1.00	0.00
ATOM	12	CA	GLY	A	12	16.210	46.133	23.554	1.00	[... truncated]

Warning in basename(x): expanded path length 17340 would be too long for

ATOM	1	CA	MET	A	1	26.318	52.948	39.518	1.00	0.00
ATOM	2	CA	ARG	A	2	27.607	49.903	37.679	1.00	0.00
ATOM	3	CA	ILE	A	3	25.007	47.987	35.714	1.00	0.00
ATOM	4	CA	ILE	A	4	25.025	44.972	33.400	1.00	0.00
ATOM	5	CA	LEU	A	5	22.465	44.494	30.650	1.00	0.00
ATOM	6	CA	LEU	A	6	22.143	40.787	29.919	1.00	0.00
ATOM	7	CA	GLY	A	7	20.072	39.724	26.945	1.00	0.00

ATOM	8	CA	ALA	A	8	19.966	38.397	23.396	1.00	0.00
ATOM	9	CA	PRO	A	9	20.372	40.256	20.073	1.00	0.00
ATOM	10	CA	GLY	A	10	17.506	42.659	19.351	1.00	0.00
ATOM	11	CA	ALA	A	11	16.395	42.374	23.007	1.00	0.00
ATOM	12	CA	GLY	A	12	16.210	46.133	23.554	1.00	[... truncated]

Warning in basename(x): expanded path length 17340 would be too long for

ATOM	1	CA	MET	A	1	26.116	53.012	39.537	1.00	0.00
ATOM	2	CA	ARG	A	2	27.613	49.804	38.094	1.00	0.00
ATOM	3	CA	ILE	A	3	25.113	47.534	36.323	1.00	0.00
ATOM	4	CA	ILE	A	4	25.178	44.906	33.567	1.00	0.00
ATOM	5	CA	LEU	A	5	22.528	44.573	30.857	1.00	0.00
ATOM	6	CA	LEU	A	6	21.910	40.963	29.825	1.00	0.00
ATOM	7	CA	GLY	A	7	19.562	39.733	27.134	1.00	0.00
ATOM	8	CA	ALA	A	8	19.198	37.936	23.799	1.00	0.00
ATOM	9	CA	PRO	A	9	20.117	39.873	20.633	1.00	0.00
ATOM	10	CA	GLY	A	10	17.308	42.287	19.870	1.00	0.00
ATOM	11	CA	ALA	A	11	15.833	42.247	23.381	1.00	0.00
ATOM	12	CA	GLY	A	12	16.130	46.042	23.507	1.00	[... truncated]

Warning in basename(x): expanded path length 17340 would be too long for

ATOM	1	CA	MET	A	1	26.116	53.012	39.537	1.00	0.00
ATOM	2	CA	ARG	A	2	27.613	49.804	38.094	1.00	0.00
ATOM	3	CA	ILE	A	3	25.113	47.534	36.323	1.00	0.00
ATOM	4	CA	ILE	A	4	25.178	44.906	33.567	1.00	0.00
ATOM	5	CA	LEU	A	5	22.528	44.573	30.857	1.00	0.00
ATOM	6	CA	LEU	A	6	21.910	40.963	29.825	1.00	0.00
ATOM	7	CA	GLY	A	7	19.562	39.733	27.134	1.00	0.00
ATOM	8	CA	ALA	A	8	19.198	37.936	23.799	1.00	0.00
ATOM	9	CA	PRO	A	9	20.117	39.873	20.633	1.00	0.00
ATOM	10	CA	GLY	A	10	17.308	42.287	19.870	1.00	0.00
ATOM	11	CA	ALA	A	11	15.833	42.247	23.381	1.00	0.00
ATOM	12	CA	GLY	A	12	16.130	46.042	23.507	1.00	[... truncated]

Warning in basename(x): expanded path length 17340 would be too long for

ATOM	1	CA	MET	A	1	26.314	52.908	40.159	1.00	0.00
ATOM	2	CA	ARG	A	2	27.545	49.868	38.192	1.00	0.00
ATOM	3	CA	ILE	A	3	25.068	47.968	35.948	1.00	0.00
ATOM	4	CA	ILE	A	4	25.151	45.033	33.515	1.00	0.00
ATOM	5	CA	LEU	A	5	22.470	44.392	30.788	1.00	0.00
ATOM	6	CA	LEU	A	6	21.943	40.686	29.878	1.00	0.00
ATOM	7	CA	GLY	A	7	19.631	39.343	27.144	1.00	0.00
ATOM	8	CA	ALA	A	8	19.541	37.811	23.671	1.00	0.00

ATOM	9	CA	PRO	A	9	20.708	39.972	20.704	1.00	0.00
ATOM	10	CA	GLY	A	10	17.767	42.213	19.662	1.00	0.00
ATOM	11	CA	ALA	A	11	16.129	42.197	23.080	1.00	0.00
ATOM	12	CA	GLY	A	12	16.430	46.007	23.217	1.00	[... truncated]

Warning in basename(x): expanded path length 17340 would be too long for

ATOM	1	CA	MET	A	1	26.314	52.908	40.159	1.00	0.00
ATOM	2	CA	ARG	A	2	27.545	49.868	38.192	1.00	0.00
ATOM	3	CA	ILE	A	3	25.068	47.968	35.948	1.00	0.00
ATOM	4	CA	ILE	A	4	25.151	45.033	33.515	1.00	0.00
ATOM	5	CA	LEU	A	5	22.470	44.392	30.788	1.00	0.00
ATOM	6	CA	LEU	A	6	21.943	40.686	29.878	1.00	0.00
ATOM	7	CA	GLY	A	7	19.631	39.343	27.144	1.00	0.00
ATOM	8	CA	ALA	A	8	19.541	37.811	23.671	1.00	0.00
ATOM	9	CA	PRO	A	9	20.708	39.972	20.704	1.00	0.00
ATOM	10	CA	GLY	A	10	17.767	42.213	19.662	1.00	0.00
ATOM	11	CA	ALA	A	11	16.129	42.197	23.080	1.00	0.00
ATOM	12	CA	GLY	A	12	16.430	46.007	23.217	1.00	[... truncated]

Warning in basename(x): expanded path length 15153 would be too long for

ATOM	1	CA	MET	A	1	26.897	52.651	40.386	1.00	0.00
ATOM	2	CA	ARG	A	2	27.764	49.793	38.108	1.00	0.00
ATOM	3	CA	ILE	A	3	25.108	48.043	36.066	1.00	0.00
ATOM	4	CA	ILE	A	4	25.145	45.145	33.625	1.00	0.00
ATOM	5	CA	LEU	A	5	22.334	44.809	31.004	1.00	0.00
ATOM	6	CA	LEU	A	6	21.823	41.124	29.893	1.00	0.00
ATOM	7	CA	GLY	A	7	19.616	39.709	27.109	1.00	0.00
ATOM	8	CA	ALA	A	8	19.563	38.665	23.499	1.00	0.00
ATOM	9	CA	PRO	A	9	20.693	41.033	20.778	1.00	0.00
ATOM	10	CA	GLY	A	10	17.799	43.303	19.832	1.00	0.00
ATOM	11	CA	ALA	A	11	16.141	42.856	23.223	1.00	0.00
ATOM	12	CA	GLY	A	12	16.070	46.600	23.736	1.00	[... truncated]

Warning in basename(x): expanded path length 15153 would be too long for

ATOM	1	CA	MET	A	1	26.897	52.651	40.386	1.00	0.00
ATOM	2	CA	ARG	A	2	27.764	49.793	38.108	1.00	0.00
ATOM	3	CA	ILE	A	3	25.108	48.043	36.066	1.00	0.00
ATOM	4	CA	ILE	A	4	25.145	45.145	33.625	1.00	0.00
ATOM	5	CA	LEU	A	5	22.334	44.809	31.004	1.00	0.00
ATOM	6	CA	LEU	A	6	21.823	41.124	29.893	1.00	0.00
ATOM	7	CA	GLY	A	7	19.616	39.709	27.109	1.00	0.00
ATOM	8	CA	ALA	A	8	19.563	38.665	23.499	1.00	0.00
ATOM	9	CA	PRO	A	9	20.693	41.033	20.778	1.00	0.00
ATOM	10	CA	GLY	A	10	17.799	43.303	19.832	1.00	0.00

ATOM	11	CA	ALA	A	11	16.141	42.856	23.223	1.00	0.00
ATOM	12	CA	GLY	A	12	16.070	46.600	23.736	1.00	[... truncated]

Warning in basename(x): expanded path length 17340 would be too long for

ATOM	1	CA	MET	A	1	26.171	53.242	39.807	1.00	0.00
ATOM	2	CA	ARG	A	2	27.395	50.196	37.826	1.00	0.00
ATOM	3	CA	ILE	A	3	24.961	48.167	35.752	1.00	0.00
ATOM	4	CA	ILE	A	4	25.131	45.119	33.464	1.00	0.00
ATOM	5	CA	LEU	A	5	22.439	44.551	30.826	1.00	0.00
ATOM	6	CA	LEU	A	6	21.829	40.965	29.707	1.00	0.00
ATOM	7	CA	GLY	A	7	19.575	39.590	26.955	1.00	0.00
ATOM	8	CA	ALA	A	8	19.497	38.196	23.410	1.00	0.00
ATOM	9	CA	PRO	A	9	20.888	40.373	20.610	1.00	0.00
ATOM	10	CA	GLY	A	10	17.925	42.582	19.676	1.00	0.00
ATOM	11	CA	ALA	A	11	16.274	42.256	23.102	1.00	0.00
ATOM	12	CA	GLY	A	12	16.317	46.035	23.623	1.00	[... truncated]

Warning in basename(x): expanded path length 17340 would be too long for

ATOM	1	CA	MET	A	1	26.171	53.242	39.807	1.00	0.00
ATOM	2	CA	ARG	A	2	27.395	50.196	37.826	1.00	0.00
ATOM	3	CA	ILE	A	3	24.961	48.167	35.752	1.00	0.00
ATOM	4	CA	ILE	A	4	25.131	45.119	33.464	1.00	0.00
ATOM	5	CA	LEU	A	5	22.439	44.551	30.826	1.00	0.00
ATOM	6	CA	LEU	A	6	21.829	40.965	29.707	1.00	0.00
ATOM	7	CA	GLY	A	7	19.575	39.590	26.955	1.00	0.00
ATOM	8	CA	ALA	A	8	19.497	38.196	23.410	1.00	0.00
ATOM	9	CA	PRO	A	9	20.888	40.373	20.610	1.00	0.00
ATOM	10	CA	GLY	A	10	17.925	42.582	19.676	1.00	0.00
ATOM	11	CA	ALA	A	11	16.274	42.256	23.102	1.00	0.00
ATOM	12	CA	GLY	A	12	16.317	46.035	23.623	1.00	[... truncated]

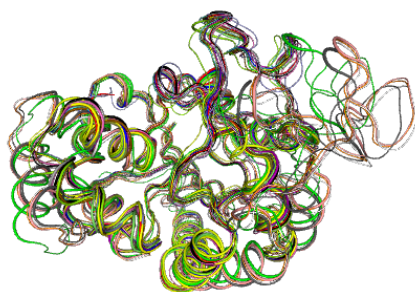
Warning in basename(x): expanded path length 17502 would be too long for

ATOM	1	CA	ASN	A	-1	29.003	55.923	45.386	1.00	0.00
ATOM	2	CA	ALA	A	0	29.542	54.100	42.093	1.00	0.00
ATOM	3	CA	MSE	A	1	27.327	52.137	39.719	1.00	0.00
ATOM	4	CA	ARG	A	2	28.229	49.294	37.431	1.00	0.00
ATOM	5	CA	ILE	A	3	25.693	47.243	35.467	1.00	0.00
ATOM	6	CA	ILE	A	4	25.609	44.433	32.869	1.00	0.00
ATOM	7	CA	LEU	A	5	22.708	44.425	30.484	1.00	0.00
ATOM	8	CA	LEU	A	6	21.200	41.475	28.635	1.00	0.00
ATOM	9	CA	GLY	A	7	18.241	41.206	26.326	1.00	0.00
ATOM	10	CA	ALA	A	8	17.354	39.864	22.930	1.00	0.00
ATOM	11	CA	PRO	A	9	18.395	41.182	19.490	1.00	0.00

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ATOM      12  CA  GLY  A   10      15.438  43.333  18.476  1.00 [... truncated]
Warning in basename(x): expanded path length 17502 would be too long for
ATOM       1  CA  ASN  A   -1      29.003  55.923  45.386  1.00  0.00
ATOM       2  CA  ALA  A    0      29.542  54.100  42.093  1.00  0.00
ATOM       3  CA  MSE  A    1      27.327  52.137  39.719  1.00  0.00
ATOM       4  CA  ARG  A    2      28.229  49.294  37.431  1.00  0.00
ATOM       5  CA  ILE  A    3      25.693  47.243  35.467  1.00  0.00
ATOM       6  CA  ILE  A    4      25.609  44.433  32.869  1.00  0.00
ATOM       7  CA  LEU  A    5      22.708  44.425  30.484  1.00  0.00
ATOM       8  CA  LEU  A    6      21.200  41.475  28.635  1.00  0.00
ATOM       9  CA  GLY  A    7      18.241  41.206  26.326  1.00  0.00
ATOM      10  CA  ALA  A    8      17.354  39.864  22.930  1.00  0.00
ATOM      11  CA  PRO  A    9      18.395  41.182  19.490  1.00  0.00
ATOM      12  CA  GLY  A   10      15.438  43.333  18.476  1.00 [... truncated]

```



```
# Vector containing PDB database codes
ids <- basename.pdb(pdb$id)

anno <- pdb.annotate(ids)
unique(anno$source)
```

```
[1] "Escherichia coli"
[2] "Escherichia coli K-12"
[3] "Escherichia coli 0139:H28 str. E24377A"
[4] "Escherichia coli str. K-12 substr. MDS42"
[5] "Photobacterium profundum"
[6] "Vibrio cholerae 01 biovar El Tor str. N16961"
```

```
anno
```

	structureId	chainId	macromoleculeType	chainLength	experimentalTechnique
1AKE_A	1AKE	A	Protein	214	X-
ray					
8BQF_A	8BQF	A	Protein	234	X-
ray					
4X8M_A	4X8M	A	Protein	214	X-
ray					
6S36_A	6S36	A	Protein	214	X-
ray					
9R6U_A	9R6U	A	Protein	214	X-
ray					
9R71_A	9R71	A	Protein	214	X-
ray					
8Q2B_A	8Q2B	A	Protein	214	X-
ray					
8RJ9_A	8RJ9	A	Protein	214	X-
ray					
6RZE_A	6RZE	A	Protein	214	X-
ray					
4X8H_A	4X8H	A	Protein	214	X-
ray					
3HPR_A	3HPR	A	Protein	214	X-
ray					
1E4V_A	1E4V	A	Protein	214	X-
ray					
5EJE_A	5EJE	A	Protein	214	X-
ray					

1E4Y_A	1E4Y	A	Protein	214	X-
ray					
3X2S_A	3X2S	A	Protein	214	X-
ray					
6HAP_A	6HAP	A	Protein	214	X-
ray					
6HAM_A	6HAM	A	Protein	214	X-
ray					
8PVW_A	8PVW	A	Protein	187	X-
ray					
4K46_A	4K46	A	Protein	214	X-
ray					
4NP6_A	4NP6	A	Protein	217	X-
ray					

	resolution	scopDomain	pfam
1AKE_A	2.000	Adenylate kinase	Adenylate kinase (ADK)
8BQF_A	2.050	<NA>	Adenylate kinase (ADK)
4X8M_A	2.600	<NA>	Adenylate kinase (ADK)
6S36_A	1.600	<NA>	Adenylate kinase (ADK)
9R6U_A	1.770	<NA>	Adenylate kinase (ADK)
9R71_A	1.610	<NA>	Adenylate kinase (ADK)
8Q2B_A	1.760	<NA> Adenylate kinase, active site lid (ADK_lid)	
8RJ9_A	1.590	<NA> Adenylate kinase, active site lid (ADK_lid)	
6RZE_A	1.690	<NA>	Adenylate kinase (ADK)
4X8H_A	2.500	<NA>	Adenylate kinase (ADK)
3HPR_A	2.000	<NA>	Adenylate kinase (ADK)
1E4V_A	1.850	Adenylate kinase	Adenylate kinase (ADK)
5EJE_A	1.900	<NA>	Adenylate kinase (ADK)
1E4Y_A	1.850	Adenylate kinase	Adenylate kinase (ADK)
3X2S_A	2.800	<NA>	<NA>
6HAP_A	2.700	<NA>	Adenylate kinase (ADK)
6HAM_A	2.550	<NA>	Adenylate kinase (ADK)
8PVW_A	2.000	<NA>	Adenylate kinase (ADK)
4K46_A	2.010	<NA>	Adenylate kinase (ADK)
4NP6_A	2.004	<NA>	Adenylate kinase (ADK)

	ligandId
1AKE_A	AP5
8BQF_A	AP5
4X8M_A	<NA>
6S36_A	CL (3),NA,MG (2)
9R6U_A	AP5,GOL,NA
9R71_A	AP5
8Q2B_A	AP5,S04,MPO

8RJ9_A ADP (2)
 6RZE_A NA (3),CL (2)
 4X8H_A <NA>
 3HPR_A AP5
 1E4V_A AP5
 5EJE_A AP5,CO
 1E4Y_A AP5
 3X2S_A JPY (2),AP5,MG
 6HAP_A AP5
 6HAM_A AP5
 8PVW_A AP5
 4K46_A ADP,AMP,PO4
 4NP6_A <NA>

ligandName

1AKE_A BIS(ADENOSINE)-5'-
 PENTAPHOSPHATE
 8BQF_A BIS(ADENOSINE)-5'-
 PENTAPHOSPHATE
 4X8M_A <NA>
 6S36_A CHLORIDE ION (3),SODIUM ION,MAGNESIUM ION (2)
 9R6U_A BIS(ADENOSINE)-5'-PENTAPHOSPHATE,GLYCEROL,SODIUM ION
 9R71_A BIS(ADENOSINE)-5'-
 PENTAPHOSPHATE
 8Q2B_A BIS(ADENOSINE)-5'-PENTAPHOSPHATE,SULFATE ION,3[N-MORPHOLINO]PROPANE SULFONIC ACID
 8RJ9_A ADENOSINE-5'-
 DIPHOSPHATE (2)
 6RZE_A SODIUM ION (3),CHLORIDE ION (2)
 4X8H_A <NA>
 3HPR_A BIS(ADENOSINE)-5'-
 PENTAPHOSPHATE
 1E4V_A BIS(ADENOSINE)-5'-
 PENTAPHOSPHATE
 5EJE_A BIS(ADENOSINE)-5'-PENTAPHOSPHATE,COBALT (II) ION
 1E4Y_A BIS(ADENOSINE)-5'-
 PENTAPHOSPHATE
 3X2S_A N-(pyren-1-ylmethyl)acetamide (2),BIS(ADENOSINE)-5'-PENTAPHOSPHATE,MAGNESIUM ION
 6HAP_A BIS(ADENOSINE)-5'-
 PENTAPHOSPHATE
 6HAM_A BIS(ADENOSINE)-5'-
 PENTAPHOSPHATE
 8PVW_A BIS(ADENOSINE)-5'-
 PENTAPHOSPHATE
 4K46_A ADENOSINE-5'-DIPHOSPHATE,ADENOSINE MONOPHOSPHATE,PHOSPHATE ION

4NP6_A <NA>

	source
1AKE_A	Escherichia coli
8BQF_A	Escherichia coli
4X8M_A	Escherichia coli
6S36_A	Escherichia coli
9R6U_A	Escherichia coli
9R71_A	Escherichia coli
8Q2B_A	Escherichia coli
8RJ9_A	Escherichia coli
6RZE_A	Escherichia coli
4X8H_A	Escherichia coli
3HPR_A	Escherichia coli K-12
1E4V_A	Escherichia coli
5EJE_A	Escherichia coli 0139:H28 str. E24377A
1E4Y_A	Escherichia coli
3X2S_A	Escherichia coli str. K-12 substr. MDS42
6HAP_A	Escherichia coli 0139:H28 str. E24377A
6HAM_A	Escherichia coli K-12
8PVW_A	Escherichia coli K-12
4K46_A	Photobacterium profundum
4NP6_A	Vibrio cholerae 01 biovar El Tor str. N16961

1AKE_A STRUCTURE OF THE COMPLEX BETWEEN ADENYLATE KINASE FROM ESCHERICHIA COLI AND THE INHIB

8BQF_A	
4X8M_A	
6S36_A	
9R6U_A	C
9R71_A	Crys
8Q2B_A	E. coli Adenylate Kinase variant D158A (
8RJ9_A	E. coli adenylate kinase Asp84
6RZE_A	
4X8H_A	
3HPR_A	
1E4V_A	
loop	
5EJE_A	Crys
1E4Y_A	
loop	
3X2S_A	
conjugated adenylate kinase	
6HAP_A	
6HAM_A	

4NP6 A

		citation	rObserved	rFree
1AKE_A	Muller, C.W., et al.	J Mol Biology (1992)	0.19600	NA
8BQF_A	Scheerer, D., et al.	Proc Natl Acad Sci U S A (2023)	0.22073	0.25789
4X8M_A	Kovermann, M., et al.	Nat Commun (2015)	0.24910	0.30890
6S36_A	Rogne, P., et al.	Biochemistry (2019)	0.16320	0.23560
9R6U_A	Mattsson, J., et al.	Biochemistry (2025)	NA	0.22790
9R71_A	Mattsson, J., et al.	Biochemistry (2025)	0.19600	0.24400
8Q2B_A	Nam, K., et al.	J Chem Inf Model (2024)	0.18320	0.22440
8RJ9_A	Nam, K., et al.	Sci Adv (2024)	0.15190	0.20290
6RZE_A	Rogne, P., et al.	Biochemistry (2019)	0.18650	0.23500
4X8H_A	Kovermann, M., et al.	Nat Commun (2015)	0.19610	0.28950
3HPR_A	Schrank, T.P., et al.	Proc Natl Acad Sci U S A (2009)	0.21000	0.24320
1E4V_A	Muller, C.W., et al.	Proteins (1993)	0.19600	NA
5EJE_A	Kovermann, M., et al.	Proc Natl Acad Sci U S A (2017)	0.18890	0.23580
1E4Y_A	Muller, C.W., et al.	Proteins (1993)	0.17800	NA
3X2S_A	Fujii, A., et al.	Bioconjug Chem (2015)	0.20700	0.25600
6HAP_A	Kantaev, R., et al.	J Phys Chem B (2018)	0.22630	0.27760
6HAM_A	Kantaev, R., et al.	J Phys Chem B (2018)	0.20511	0.24325
8PVW_A	Rodriguez, J.A., et al.	To be published	0.18590	0.23440
4K46_A	Cho, Y.-J., et al.	To be published	0.17000	0.22290
4NP6_A	Kim, Y., et al.	To be published	0.18800	0.22200

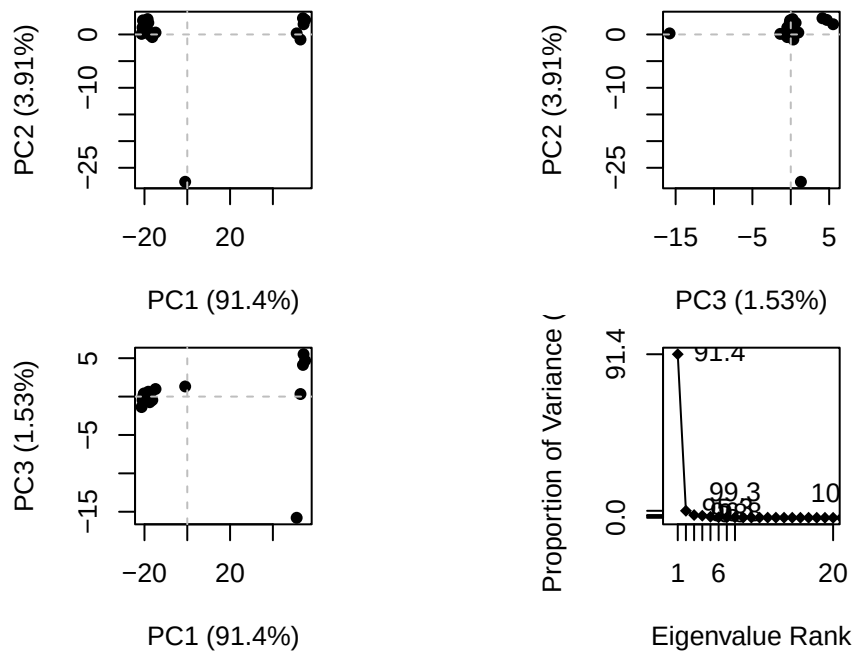
8PVW_A 0.18340 P 2 21 21

```
4K46_A 0.16730 P 21 21 21
4NP6_A 0.18600      P 43
```

Principal component analysis

PCA of all this structural data (x, y, and z atom coordinates):

```
pc <- pca(pdbbs)
plot(pc)
```

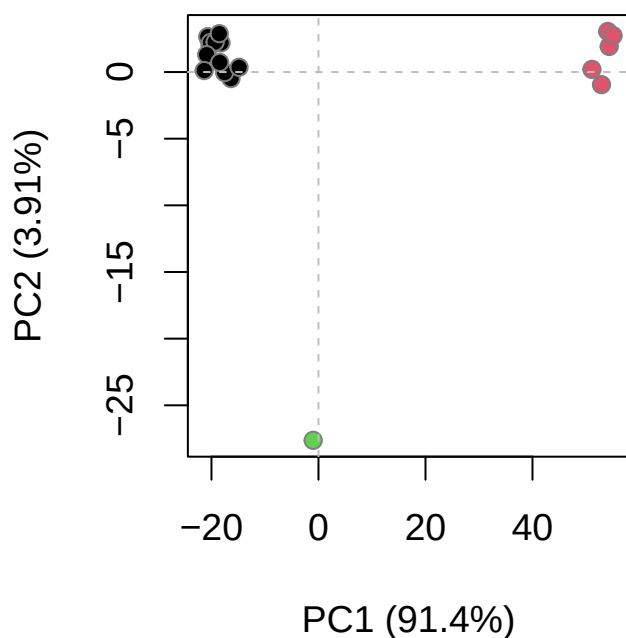


```
# Calculate RMSD
rd <- rmsd(pdbbs)
```

Warning in rmsd(pdbbs): No indices provided, using the 182 non NA positions

```
# Structure-based clustering
hc.rd <- hclust(dist(rd))
grps.rd <- cutree(hc.rd, k=3)

plot(pc, 1:2, col="grey50", bg=grps.rd, pch=21, cex=1)
```

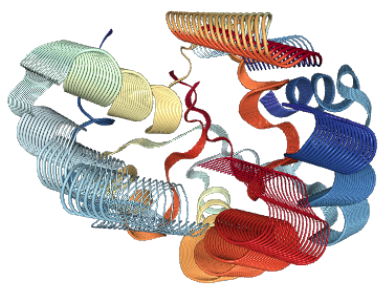


Interactive view of the PC1 captured structural differences:

```
view.pca(pc)
```

Warning in basename(x): expanded path length 501982 would be too long for

MODEL	1								
ATOM	1	CA	ALA	1	26.417	52.833	39.777	1.00	0.00
ATOM	2	CA	ALA	2	27.634	49.876	37.714	1.00	0.00
ATOM	3	CA	ALA	3	25.090	47.935	35.655	1.00	0.00
ATOM	4	CA	ALA	4	25.212	44.981	33.256	1.00	0.00
ATOM	5	CA	ALA	5	22.437	44.653	30.666	1.00	0.00
ATOM	6	CA	ALA	6	21.828	41.101	29.497	1.00	0.00
ATOM	7	CA	ALA	7	19.459	39.916	26.810	1.00	0.00
ATOM	8	CA	ALA	8	19.045	38.273	23.448	1.00	0.00
ATOM	9	CA	ALA	9	19.910	40.076	20.230	1.00	0.00
ATOM	10	CA	ALA	10	17.099	42.478	19.433	1.00	0.00
ATOM	11	CA	ALA	11	15.601	42.393	22.927	1.00	0.00
ATOM	12	CA	ALA	12	15.880	46.1	[... truncated]		



```
mktrj(pc, file = "pca.pdb")
```

```
#Plotting results with ggplot2
library(ggplot2)
library(ggrepel)

df <- data.frame(PC1=pc$z[,1],
                  PC2=pc$z[,2],
                  col=as.factor(grps.rd),
                  ids=ids)

p <- ggplot(df) +
  aes(PC1, PC2, col=col, label=ids) +
  geom_point(size=2) +
  geom_text_repel(max.overlaps = 20) +
  theme(legend.position = "none")
p
```

